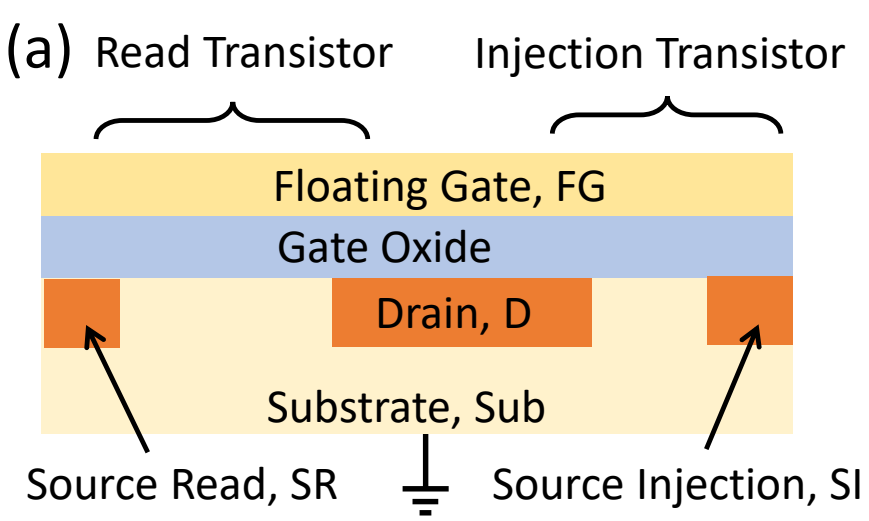
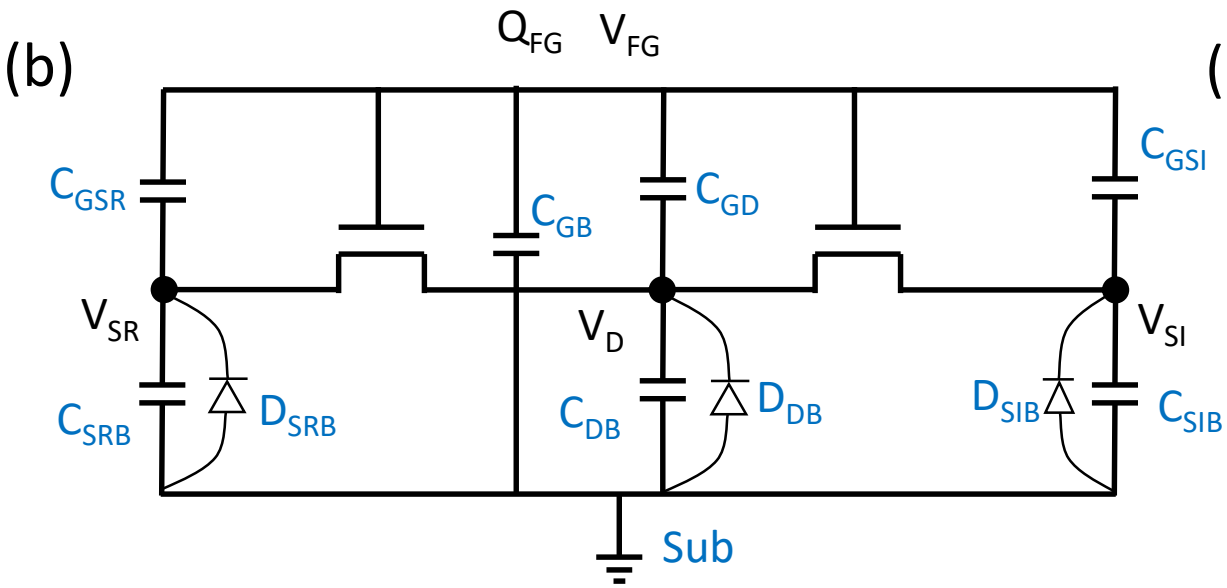
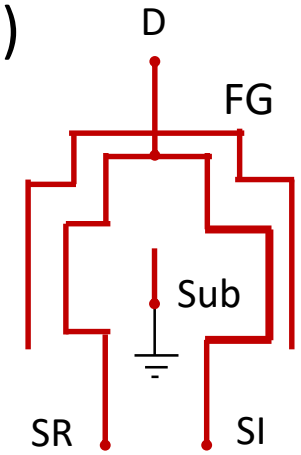
Y-Flash data analysis and device modelling

-- Wei Wang

The capacitor network for voltage coupling





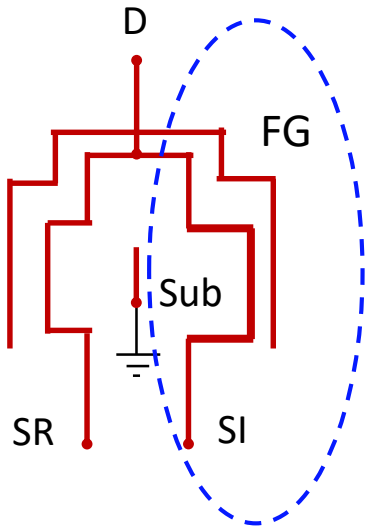


$$Q_{FG} = C_{GSR}(V_{FG} - V_{SR}) + C_{GD}(V_{FG} - V_D) + C_{GSI}(V_{FG} - V_{SI}) + C_{GB}V_{FG}$$

$$C_{GD} > C_{GSR} \approx C_{GSI}$$

$$V_{FG} = f(Q_{FG}, V_D, V_{SR}, V_{SI})$$

➢ Model for the transistors – Source/Drain/Gate current



- V_{FG}-V_{TH}<0, Subthreshold region

$$I_{S,sub} = I_{S0} e^{\frac{q(V_{FG}-V_{TH})}{nkT}} (1 - e^{-q\frac{V_D - V_S}{kT}})$$

- 0< V_{FG}-V_{TH}< $V_D - V_S$, Saturation region

$$I_{S,ab} = \frac{K}{2}(V_{FG} - V_{TH})^2$$

- V_{FG} -V_{TH}> $V_D - V_S$, Linear region

$$I_{S,ab} = K\left(V_{FG} - V_{TH} - \frac{V_D - V_S}{2}\right)(V_D - V_S)$$

$$I_S = \left(\frac{1}{I_{S,sub}^m} + \frac{1}{I_{S,ab}^m}\right)^{-\frac{1}{m}}$$

$$I_{S0} = \frac{W}{L} \propto_e C_{ox}^* \left(\frac{kT}{q}\right)^2 (n-1)$$

$$K = \frac{W}{L} \mu C_{ox}$$

- For readout transistor:

$V_{TH} = V_{TH,r}, K = K_r, I_{S0} = I_{SR0}, V_S = V_{SR}$ ⇨ I_{SR}

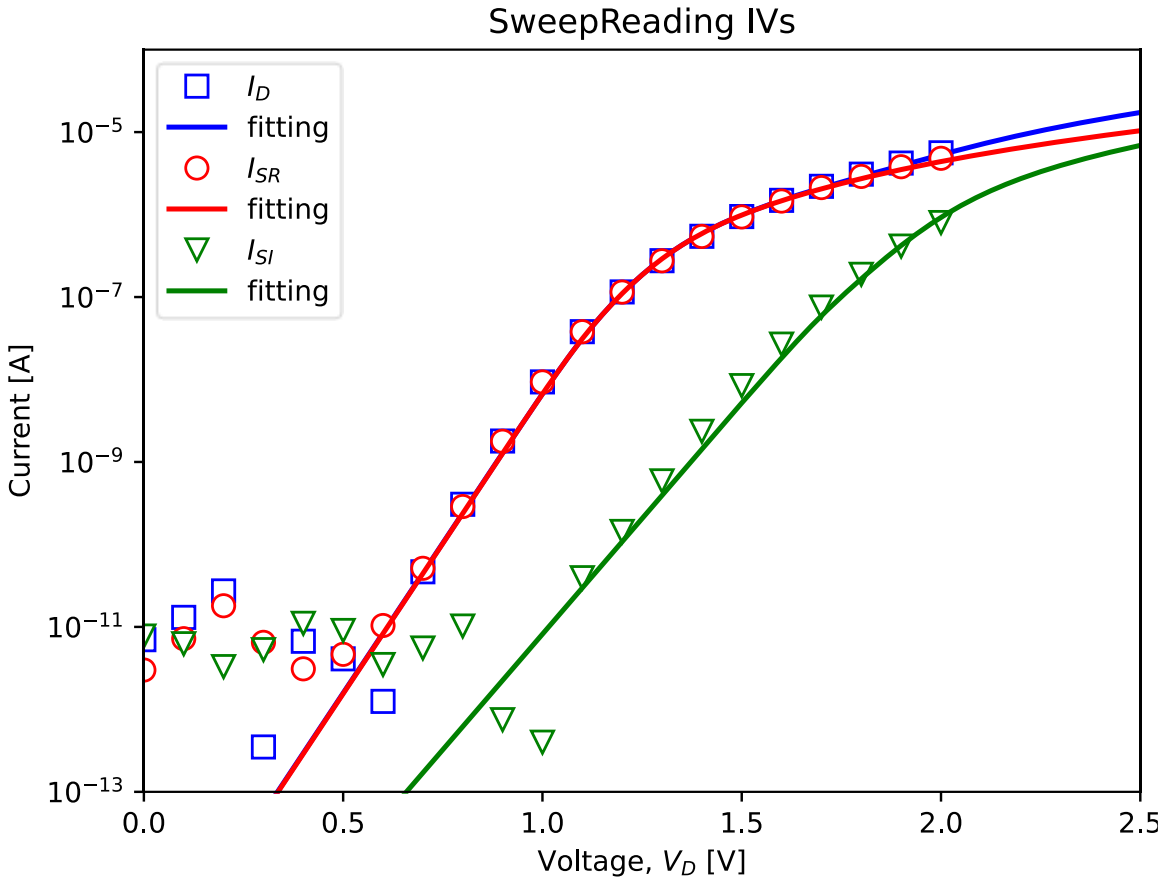
- For injection transistor:

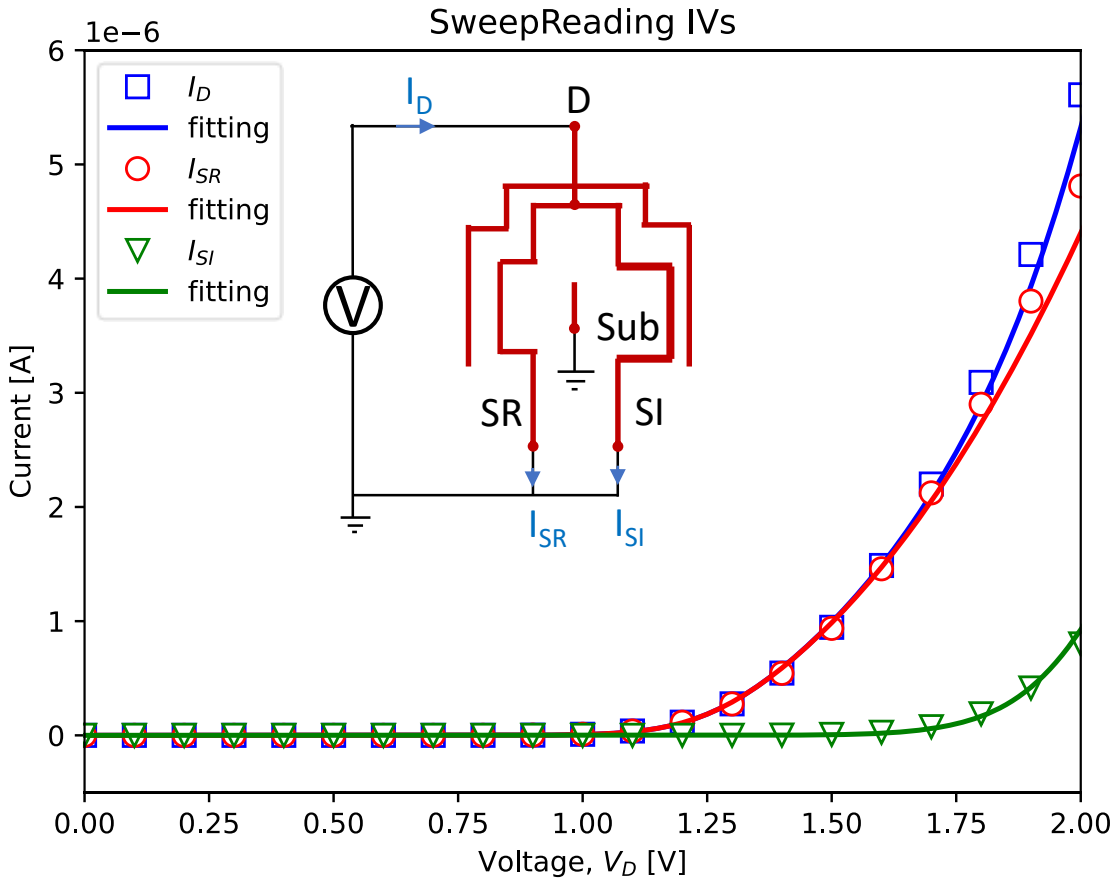
$V_{TH} = V_{TH,i}, K = K_i, I_{S0} = I_{SI0}, V_S = V_{SI}$ ⇨ I_{SI}

$$I_D = I_{SR} + I_{SI}$$

$V_{TH,r} < V_{TH,r}, K_i = 2K_r$

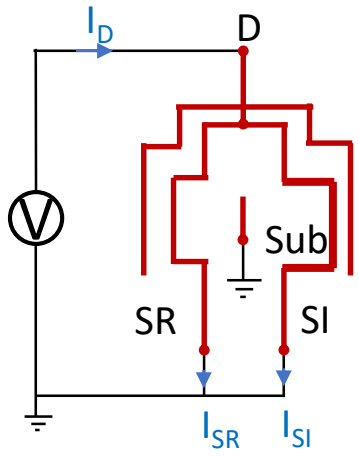
IV curves for reading in Subthreshold and Normal region



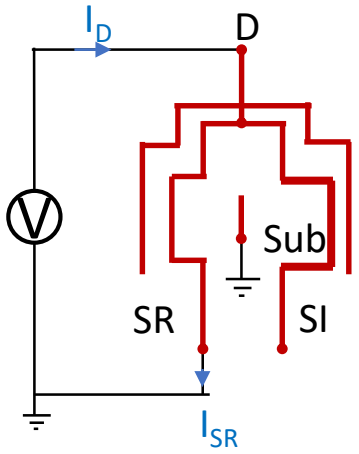


IV curves for reading: more reading modes

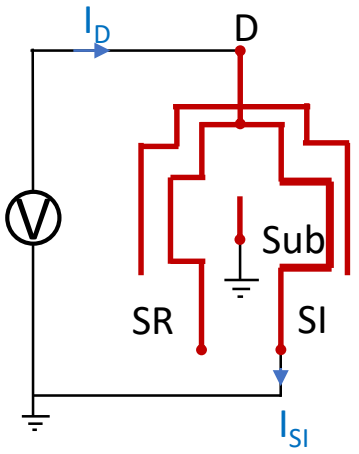
Mode 1: SR and SI shorted

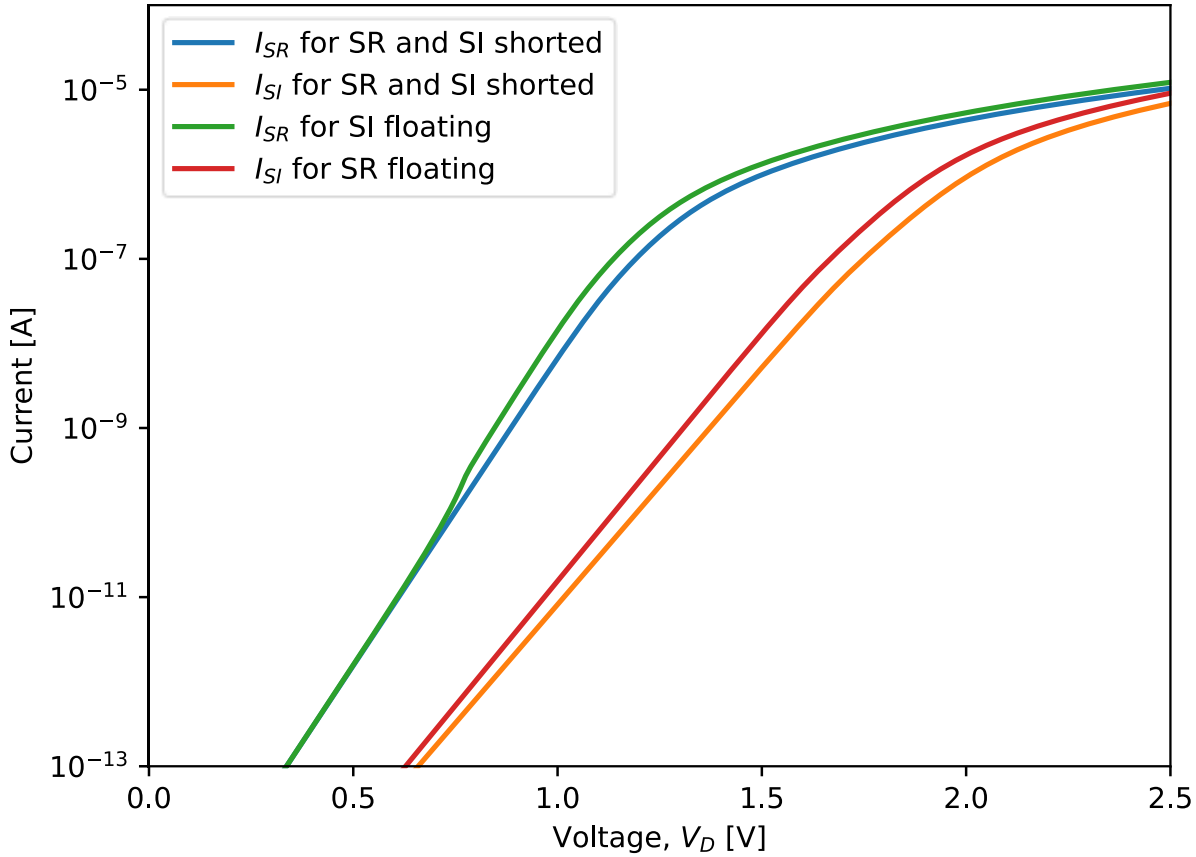


Mode 2: SI floating

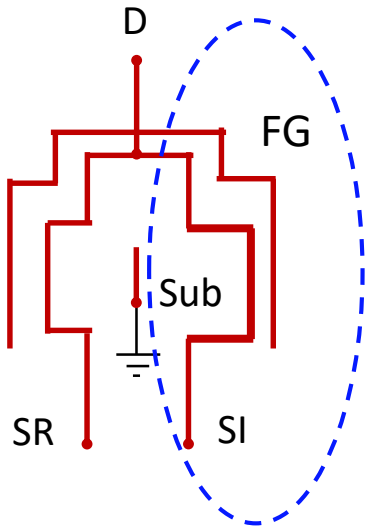


Mode 3: SI floating

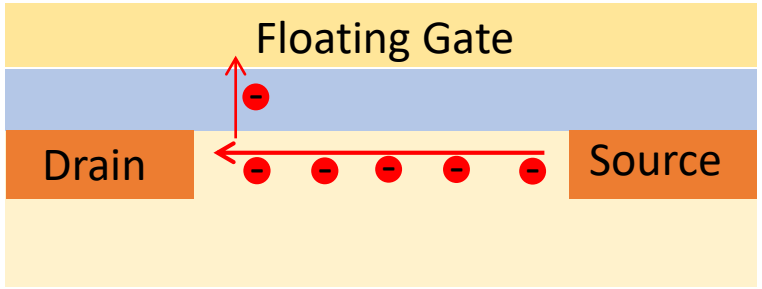




➢ Program and Erase mechanism - Current from/to the floating gate

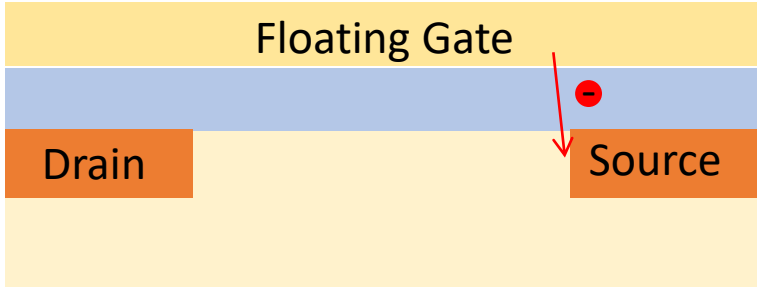


➢ Channel hot electron injection (lucky electron)



- Happens only in the injection transistor

➢ Tunnel electron



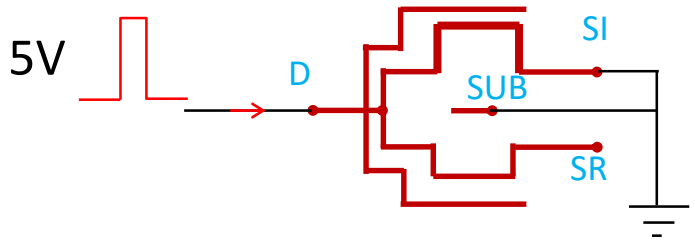
- Happens on both transistor

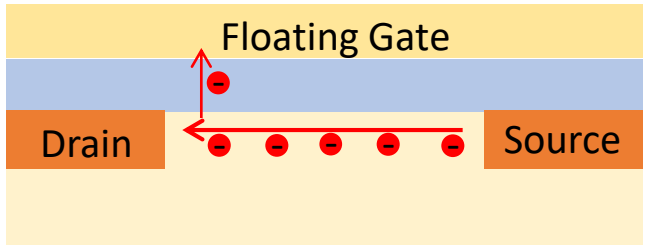
➢ Total gate current

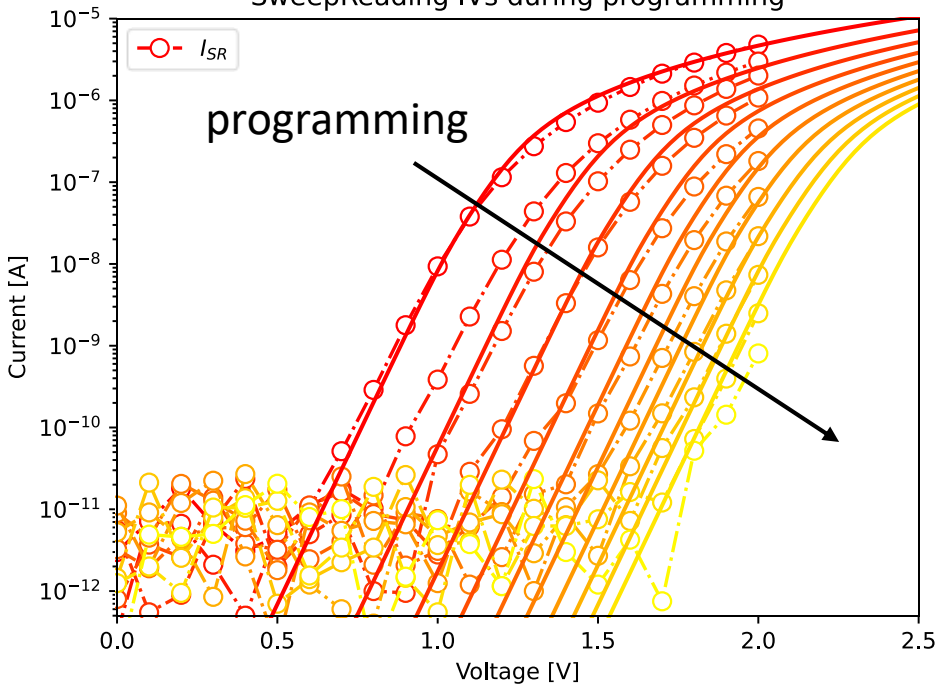
$$I_G = \underbrace{\xi(V_{FG} - V_{bi})^2 \, e^{-\frac{\beta}{V_{FG}-V_{bi}}}}_{\text{Tunnelling current}} \underbrace{- I_S P_0 e^{-\frac{V_\alpha}{V_{FG}}}}_{\text{Injection current}}$$

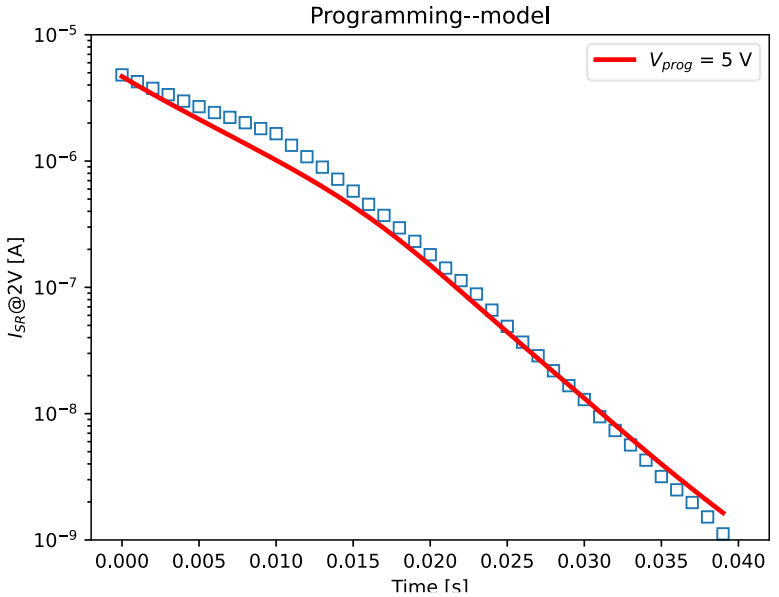
$$\frac{dQ_{FG}}{dt} = I_G$$

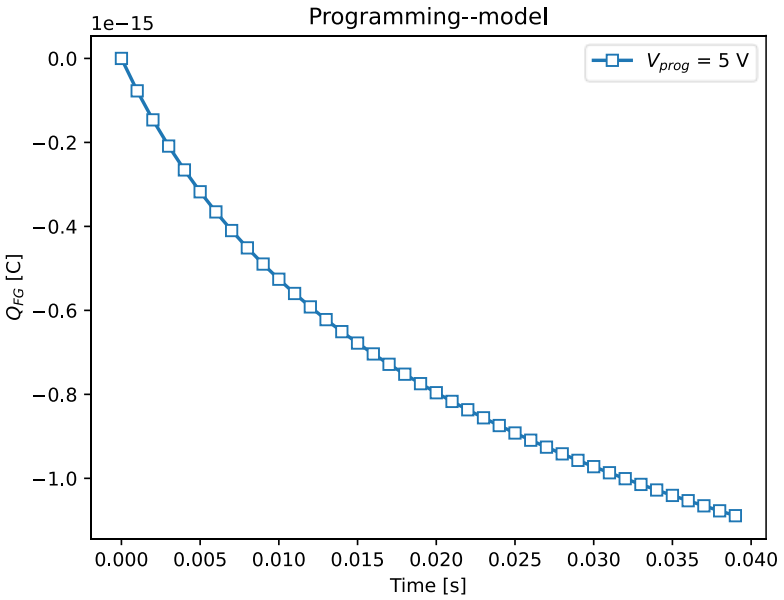
IV reading during programing



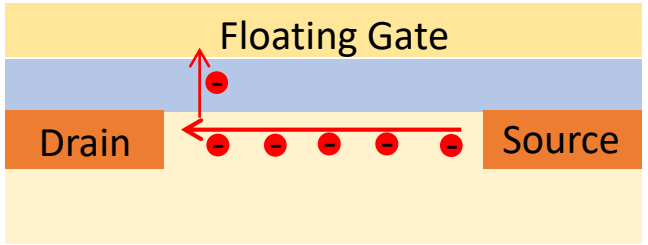




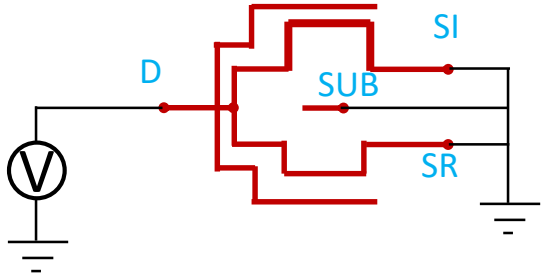




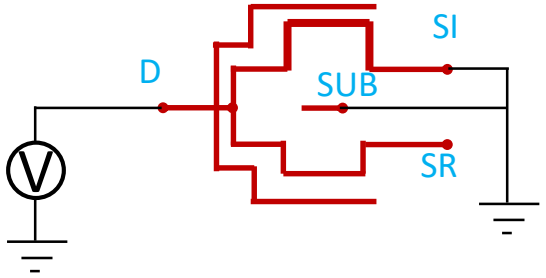
➢ More program modes

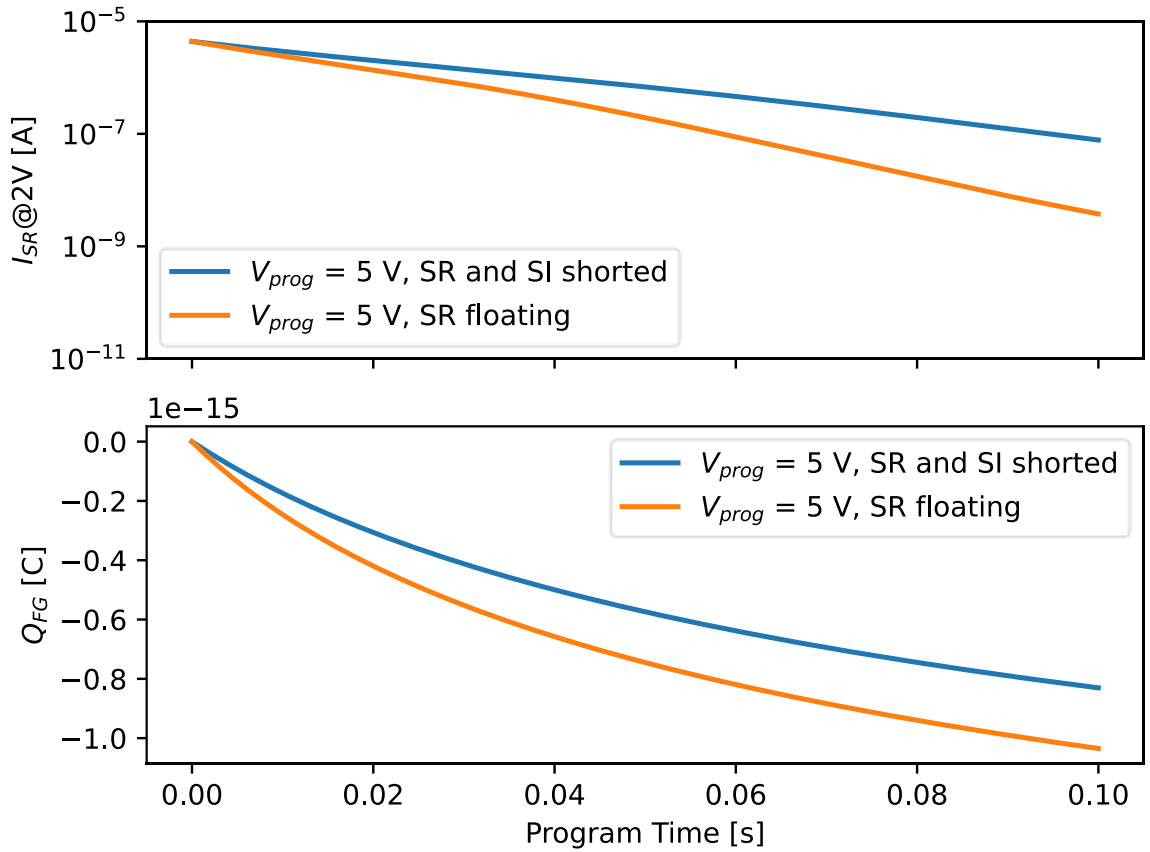


- Mode 1: SI and SR shorted



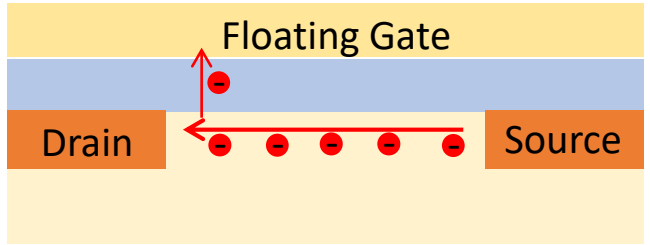
- Mode 2: SR floating



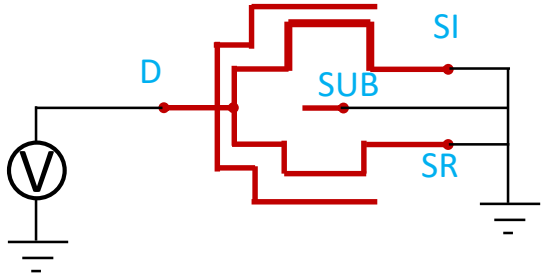


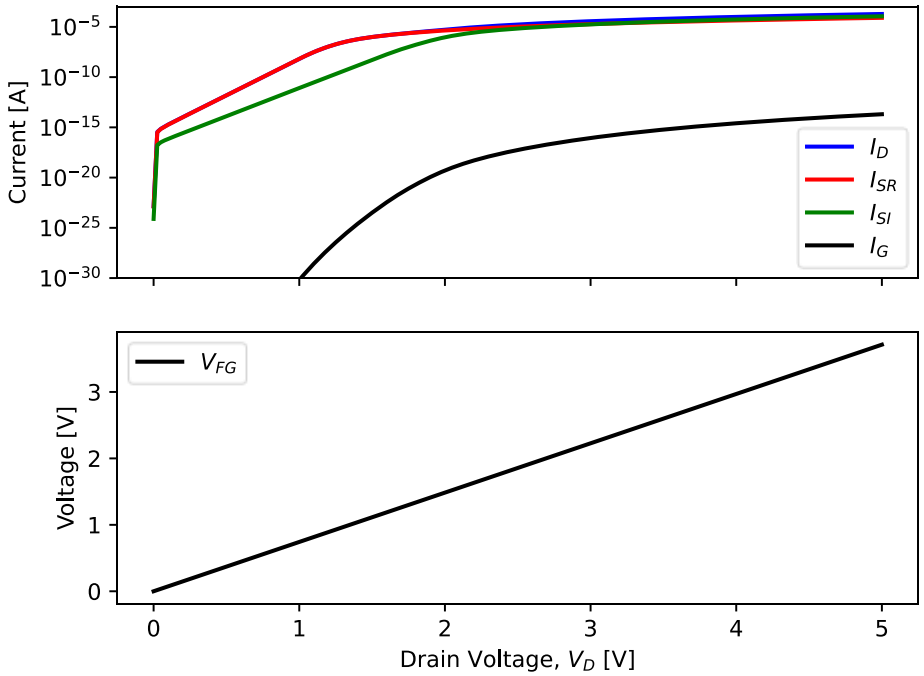
➢ Mode 2 is slightly more efficient than Mode 1

More program modes

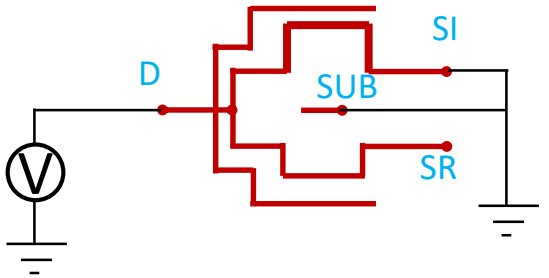


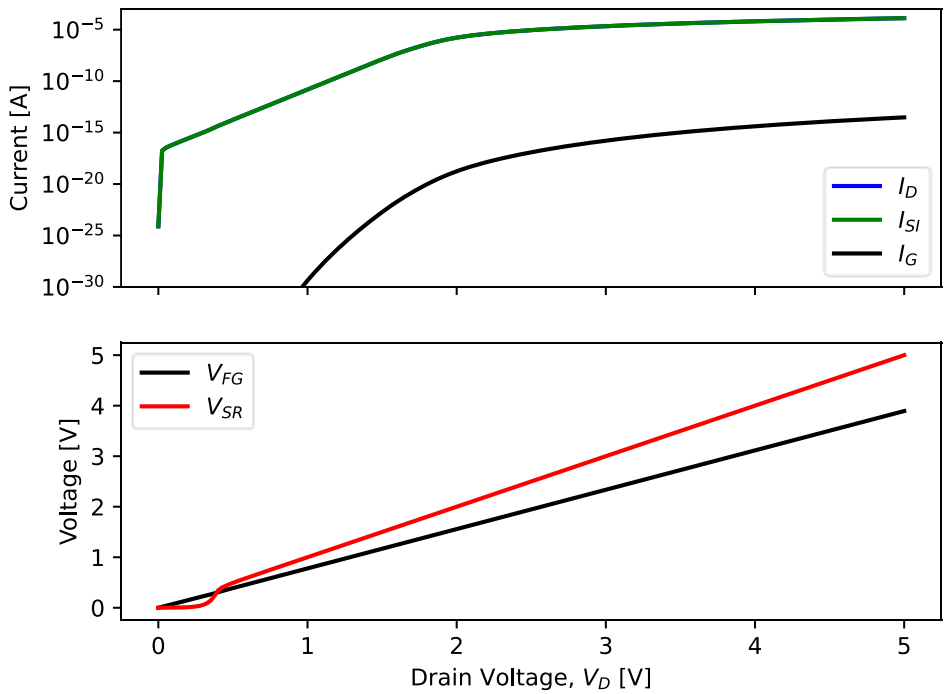
- Mode 1: SI and SR shorted





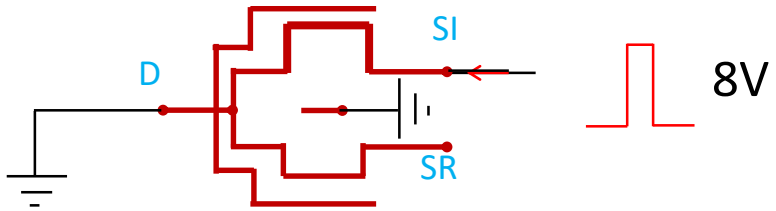
- Mode 2: SR floating

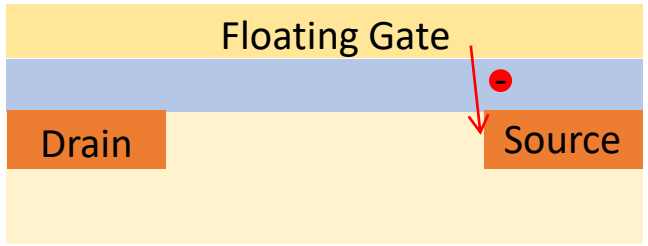


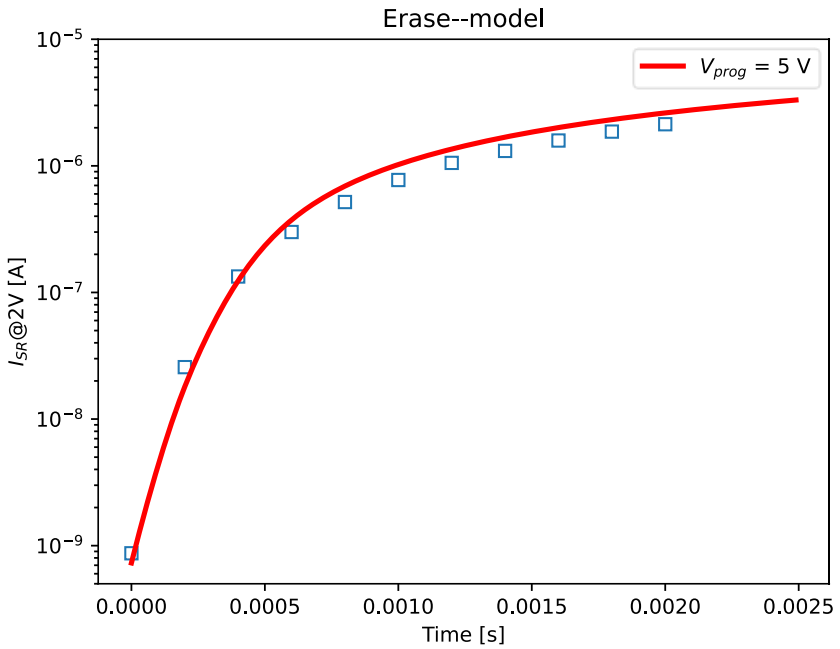


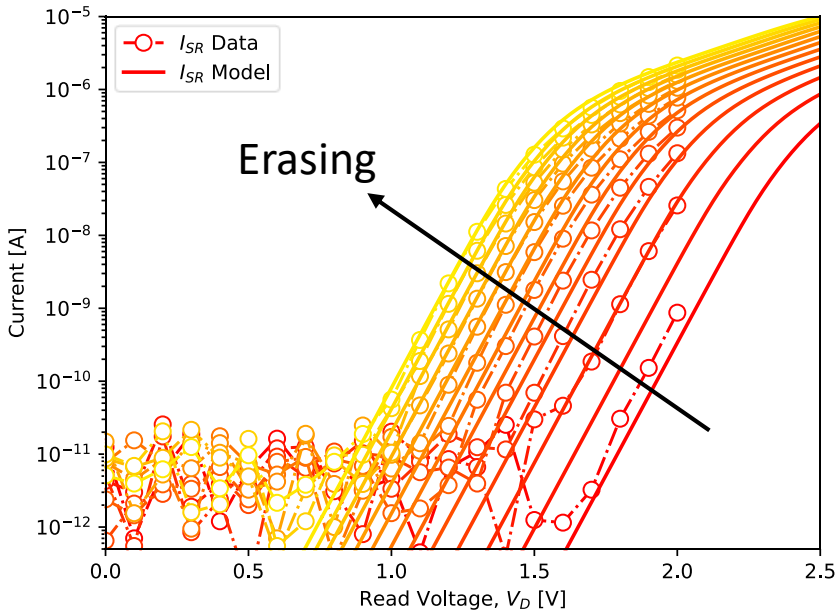
➢ Mode 2 is slightly more efficient than Mode 1

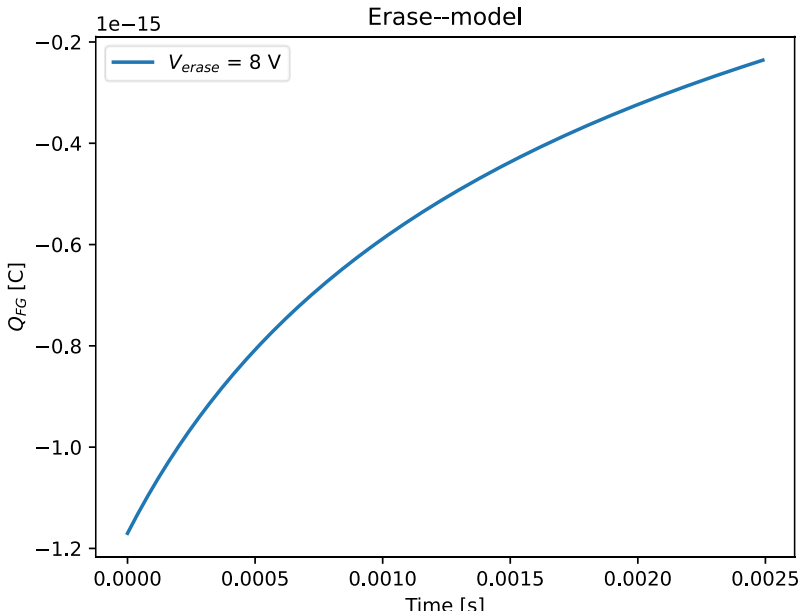
IV reading curves during erasing



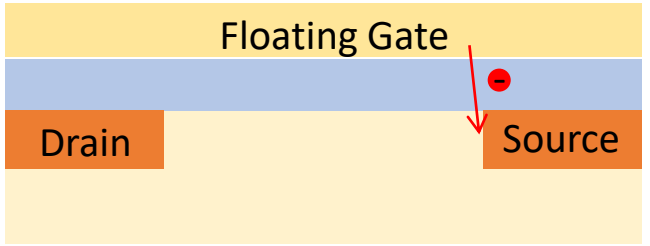




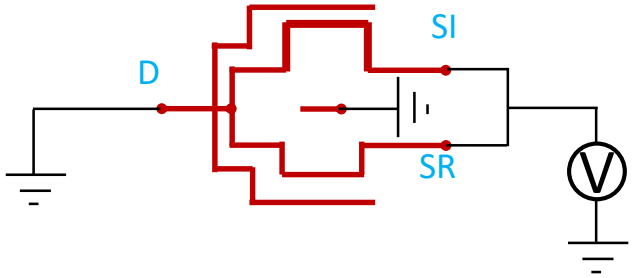


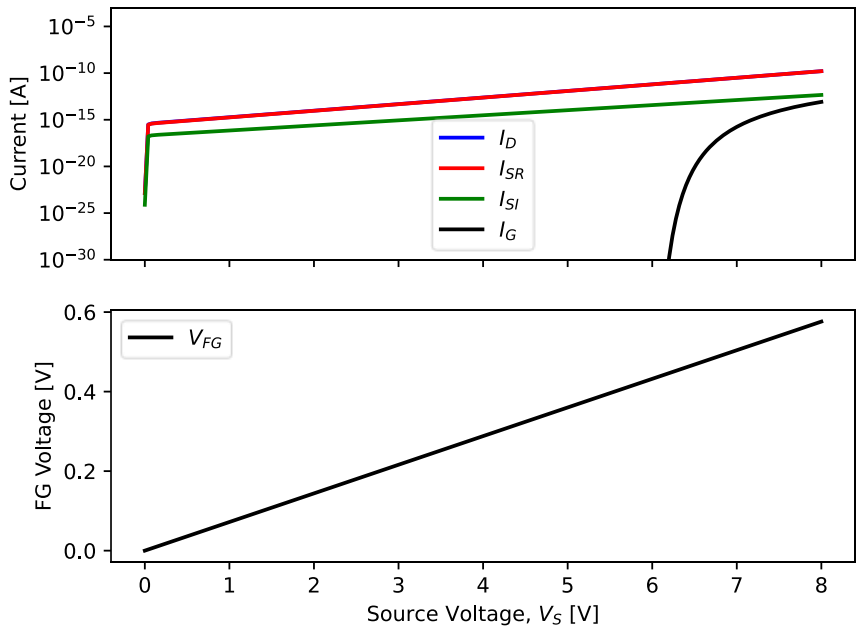


➢ More erase modes

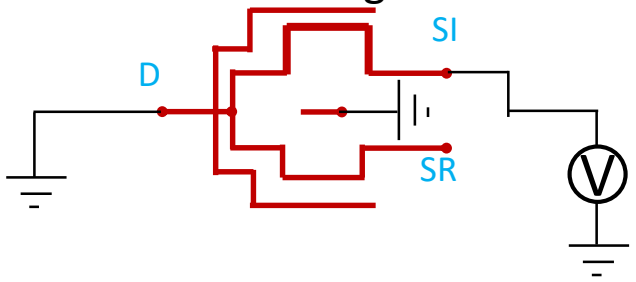


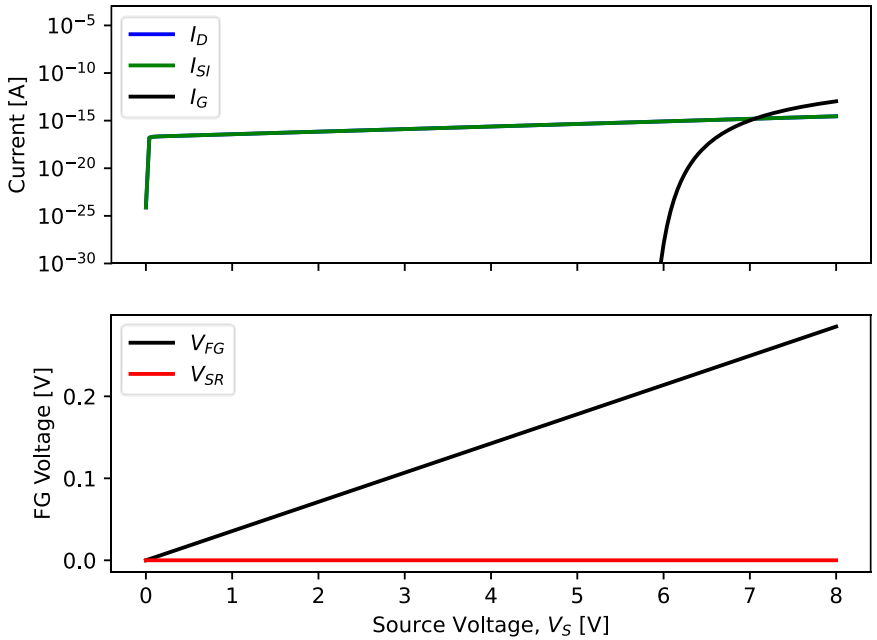
- Mode 1: SI and SR shorted





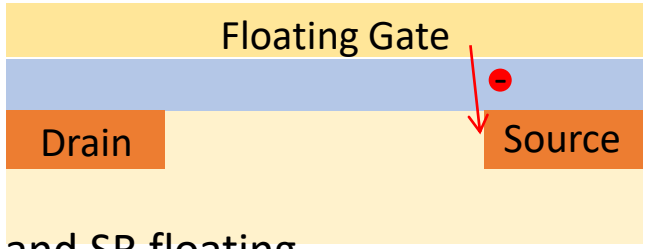
- Mode 2: SR floating



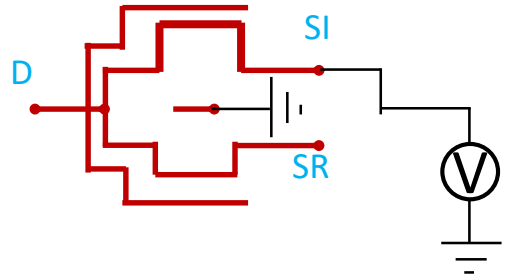


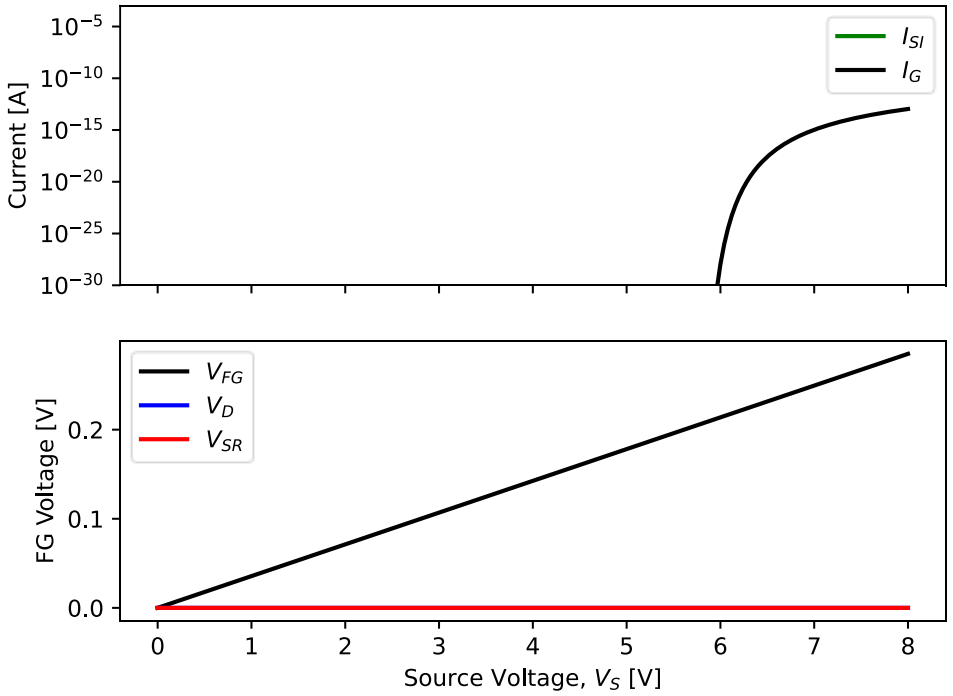
➢ Mode 2 is slightly more efficient than Mode 1

More erase modes

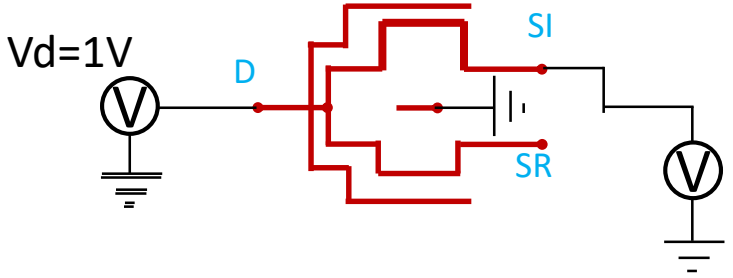


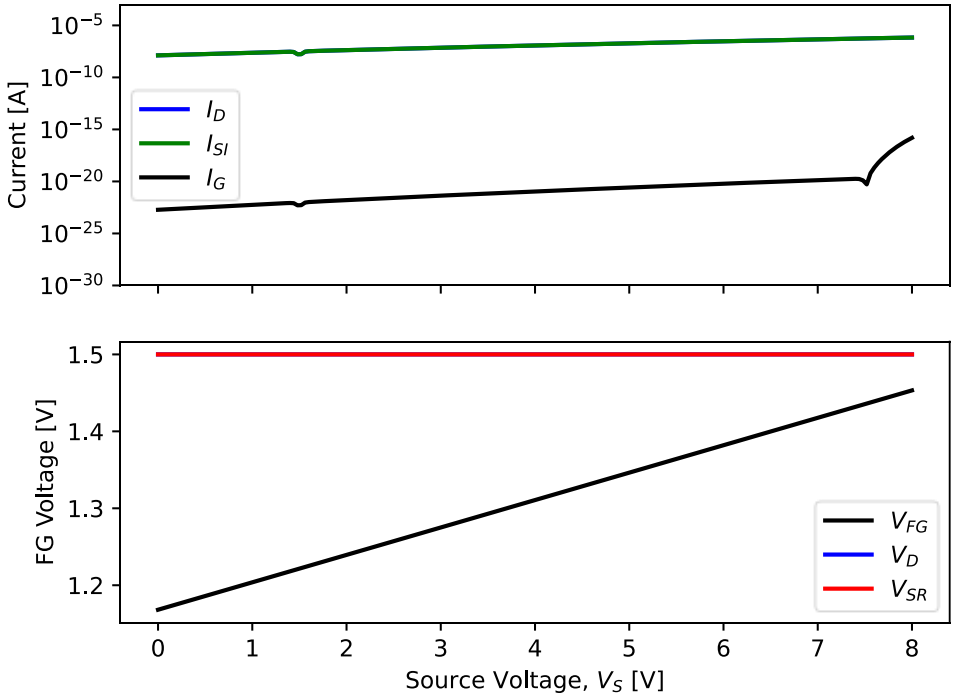
- Mode 3: D and SR floating





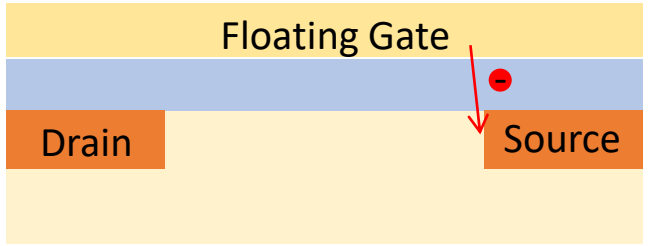
- Mode 4: A small voltage on Drain, and SR floating

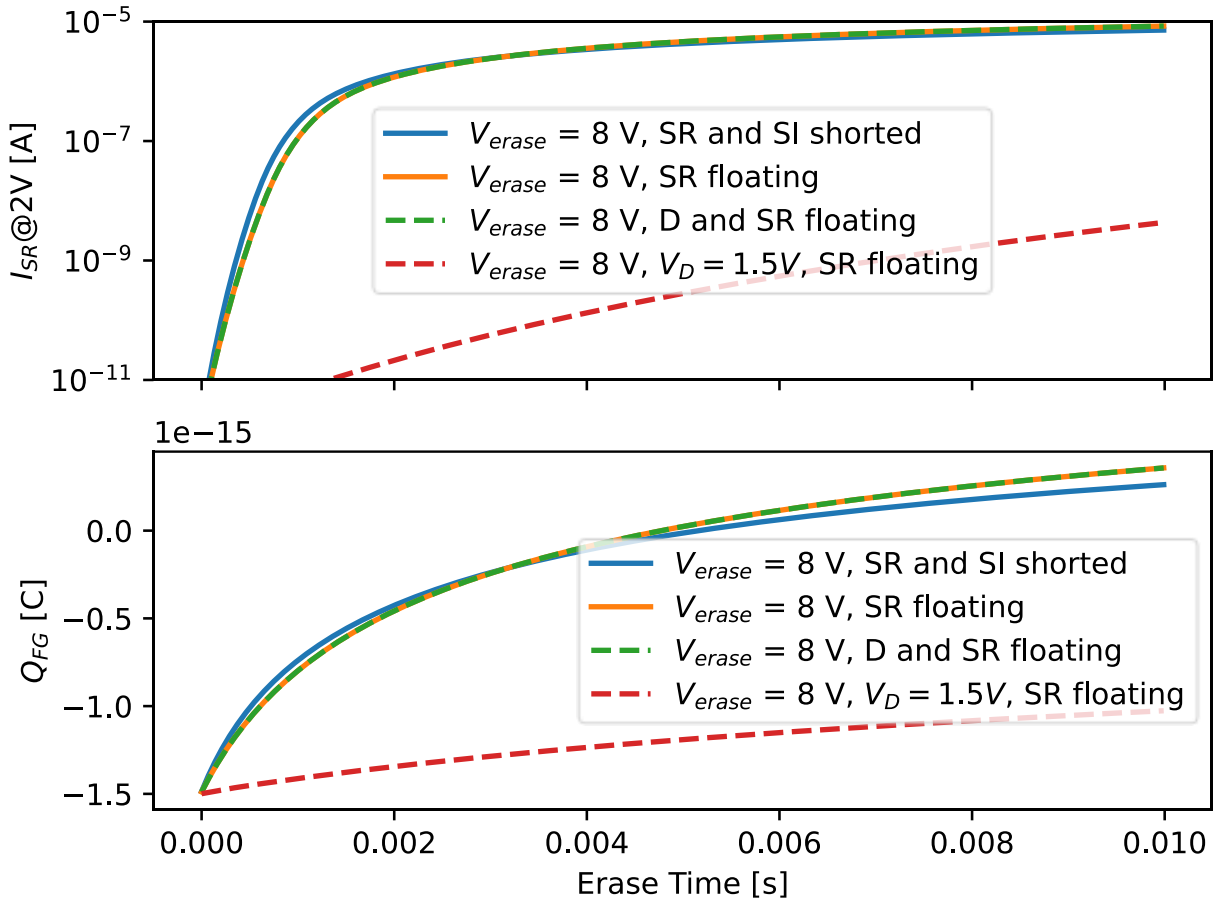




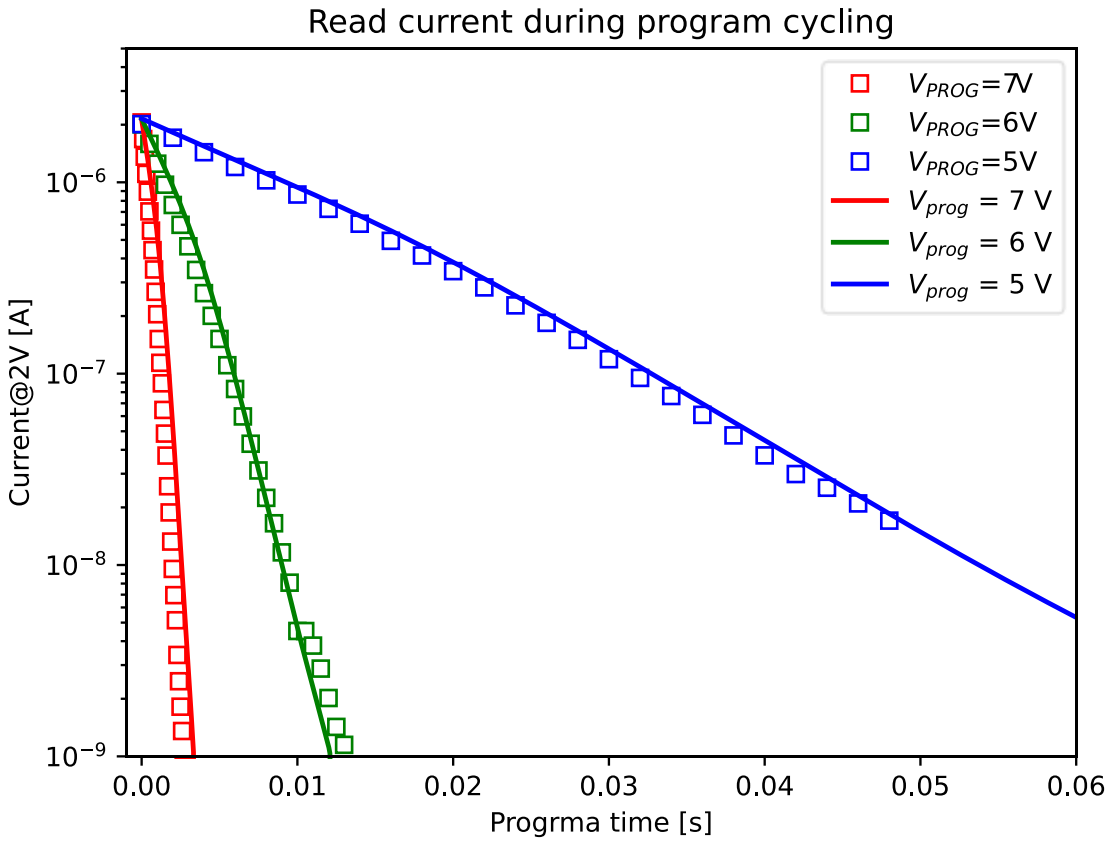
- Whether Drain is floating or grounded does not matter much
- Applying a small voltage on Drain can prevent Erase

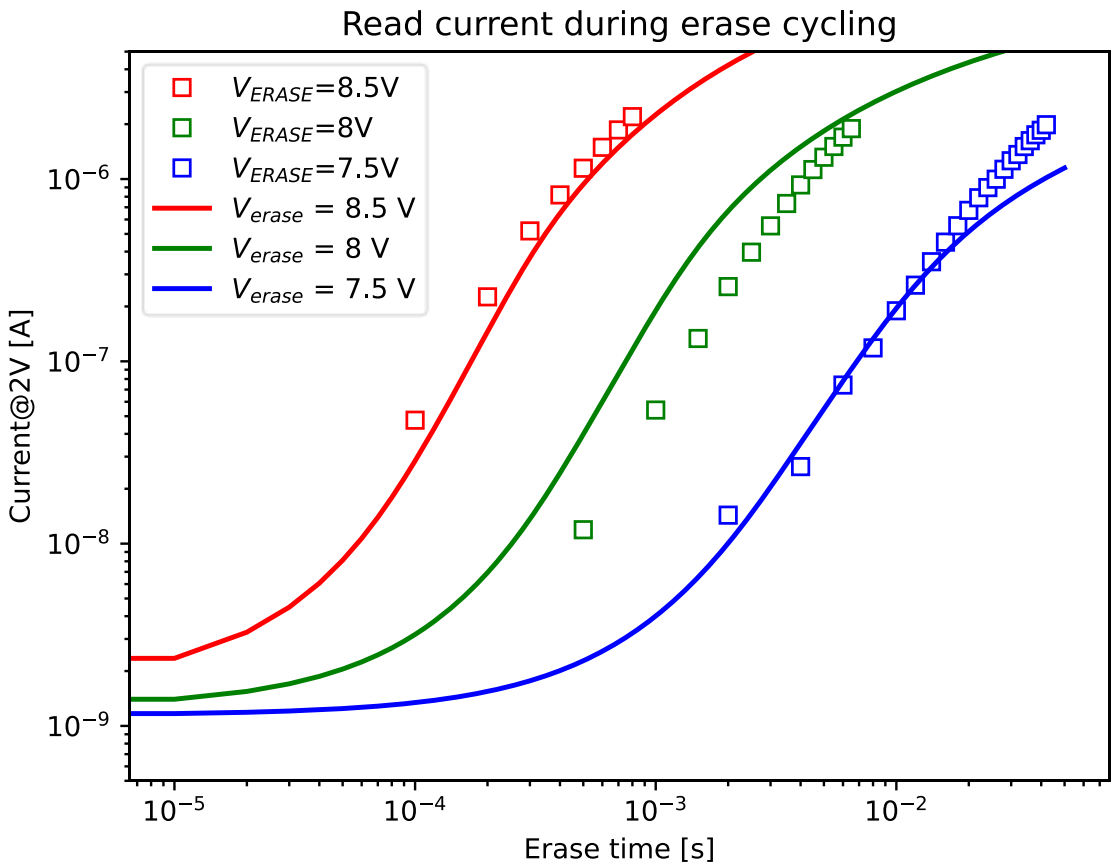
More erase modes



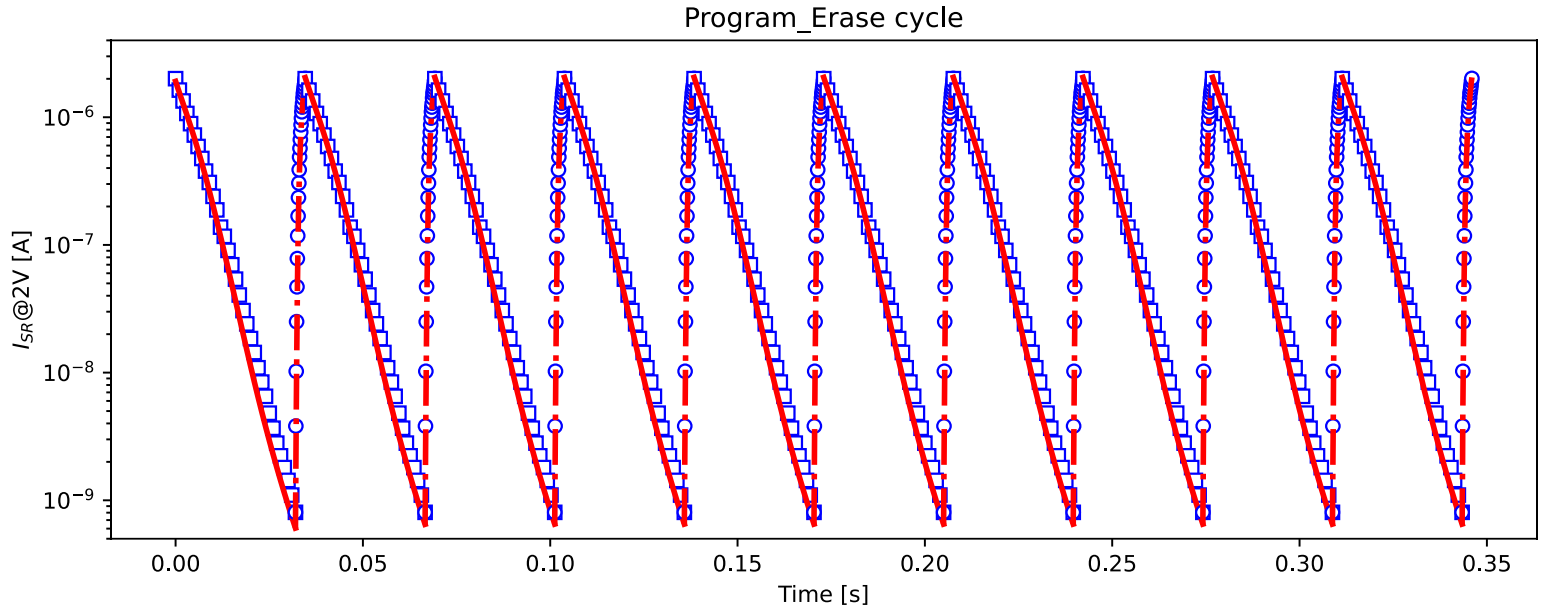


Program and erase at different voltages

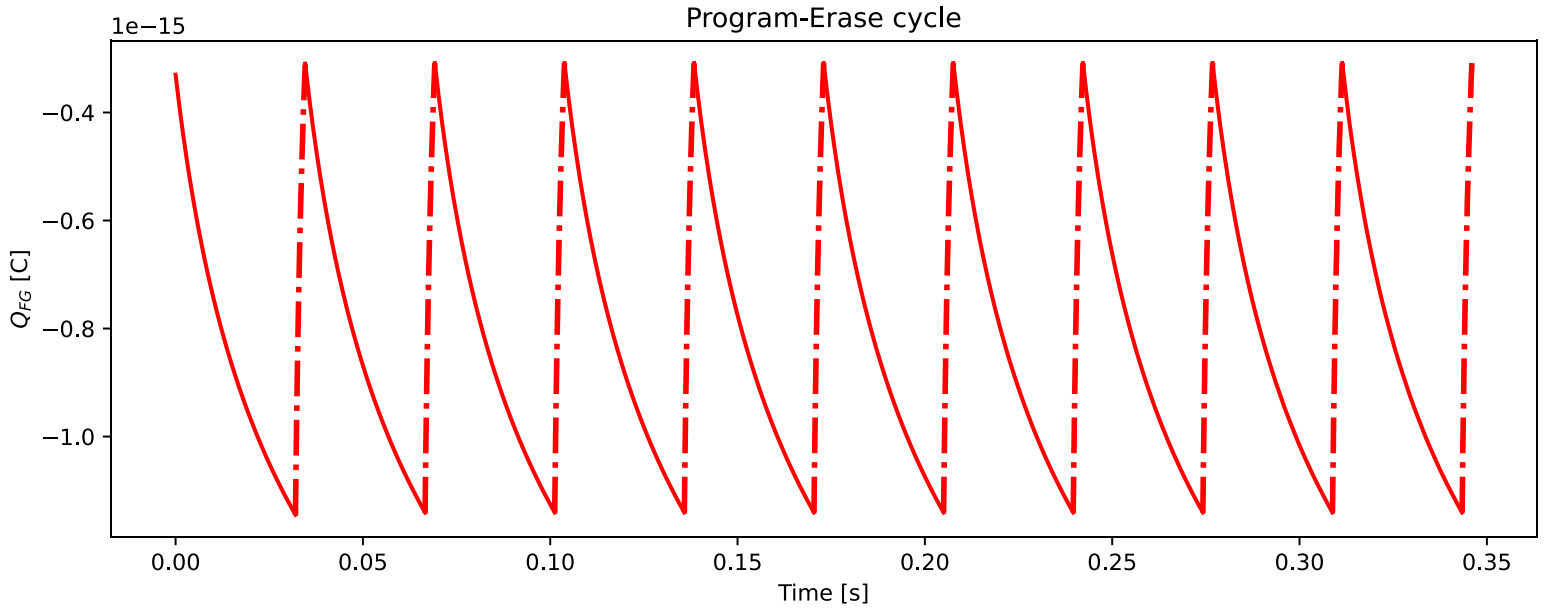




Cycling of the program and erase operations

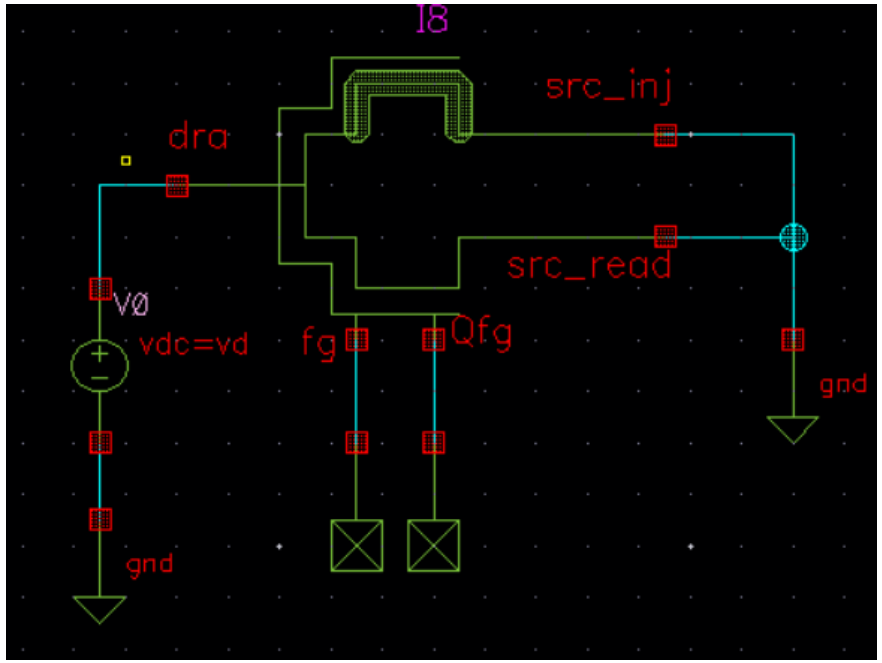


- Data vs. Model



- Up to 100 cycle data are available

➢ Verilog-a Model - 1




```
// VerilogA for Y_Flash, veriloga
// wei 20210511

`include "constants.vams"
`include "disciplines.vams"

discipline Chrg
    potential Charge;
enddiscipline

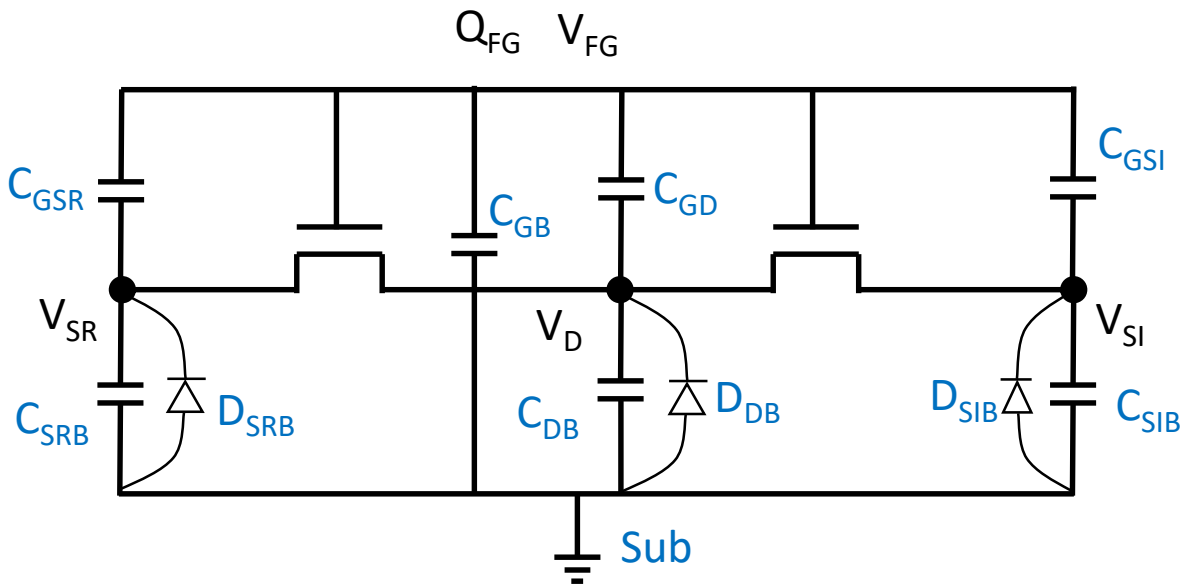
module y_flash (src_read, src_inj, dra, fg, Qfg);

    inout src_read, src_inj, dra;
    output fg, Qfg;   //outputs only for monitoring the state of the devices

    electrical src_read, src_inj, dra;
    electrical fg, fg_d; //floating gate and dummy floating gate
    Chrg Qfg;            //self defined discipline for charge in the floating gate
    ground gnd;          //substrate of the device is always grounded

    //PARAMETERS
    parameter real Qfg_ini  = 0 from [-5e-15:1e-15];
    //Parasitic Capacitors [from Loai D. Nat. Elect., 2, 596 (2019).]
    parameter real Cgd  = 1.0e-15;
    parameter real Cgsr = 0.049e-15;
    parameter real Cgsi = 0.048e-15;
    parameter real Cdb = 0.64e-15;
    parameter real Cgb = 0.24e-15;
    //Parameters for readout transistor
    parameter real r_Vth    = 0.82;
    parameter real r_K      = 1.9e-5;
    parameter real r_Is0    = 4e-8;
    parameter real r_n      = 1.7;
    parameter real r_Prob0 = 0;
    parameter real r_Va     = 0;
    parameter real r_beta   = 10;
    parameter real r_Vbi    = 5.5;
    parameter real r_xi     = 2e-12;
    //Parameters for injection transistor
    parameter real i_Vth    = 0.82+0.52;
    parameter real i_K      = 1.9e-5*2;
    parameter real i_Is0    = 4e-8*2;
    parameter real i_n      = 1.7*1.3;
    parameter real i_Prob0 = 3.9e-8;
    parameter real i_Va     = 20;
    parameter real i_beta   = 10;
```

Verilog-a Model - 2




```
    parameter real r_Va    = 0;
    parameter real r_beta  = 10;
    parameter real r_Vbi   = 5.5;
    parameter real r_xi    = 2e-12;
    //Parameters for injection transistor
    parameter real i_Vth   = 0.82+0.52;
    parameter real i_K     = 1.9e-5*2;
    parameter real i_Is0   = 4e-8*2;
    parameter real i_n     = 1.7*1.3;
    parameter real i_Prob0 = 3.9e-8;
    parameter real i_Va    = 20;
    parameter real i_beta  = 10;
    parameter real i_Vbi   = 5.5;
    parameter real i_xi    = 2.0e-12;

    ////////////////////// MAIN ////////////////////

    // Instances of the two transistors: read transistor and injection transistor
    Transistorfg #(.Vth(r_Vth), .K(r_K), .Is0(r_Is0), .n(r_n), .Prob0(r_Prob0), .Va(r_Va), .beta(r_beta), .Vbi(r_Vbi), .xi(r_xi)) Source_Read(.g(fg_d), .dra(dra), .src(src_read));
    Transistorfg #(.Vth(i_Vth), .K(i_K), .Is0(i_Is0), .n(i_n), .Prob0(i_Prob0), .Va(i_Va), .beta(i_beta), .Vbi(i_Vbi), .xi(i_xi)) Source_Inj(.g(fg_d),.dra(dra),.src(src_inj));

    // PN-Junction Diodes between the terminals and substrates (there are also parasitic capactor and resistor in the PN junctions)
    PNDiode #(.Area(1), .Cjo(Cdb))         DRdgnd(.p(gnd), .n(dra));
    PNDiode #(.Area(0.05), .Cjo(0.05*Cdb)) DRsrgnd(.p(gnd), .n(src_read));
    PNDiode #(.Area(0.05), .Cjo(0.05*Cdb)) DRsignd(.p(gnd), .n(src_inj));

    analog begin
        //the floating gate node does not following KCL. A dummy floating point was employeed to collect the gate current from the transistors (program and erase currents).
        //integrate the gate current in the floating gate
        V(fg_d) <+ V(fg);;
        Q(Qfg) <+ idt(I(fg_d), Qfg_ini);
        V(fg) <+ (Q(Qfg)+Cgsr*V(src_read,gnd)+Cgd*V(dra,gnd)+Cgsi*V(src_inj,gnd))/(Cgsr+Cgd+Cgsi+Cgb);

        //Addtional Capacitors can be seen from input terminals, affecting the transient behaviors
        I(dra,fg) <+ Cgd*ddt(V(dra,fg));
        I(src_read,fg) <+  Cgsr*ddt(V(src_read,fg));
        I(src_inj,fg) <+  Cgsi*ddt(V(src_inj,fg));

        // $monitor("Time %f ms, Qfg=%f fC, Vfg= %f V",$realtime*1e3, Q(Qfg)*1e15, V(fg));
    end
endmodule
```

➢ Verilog-a Model - 3

```
module Transistorfg(g,dra,src);
    inout g, dra, src; // gate, drain, source
    electrical g, dra, src;
    ground gnd;        // substrate

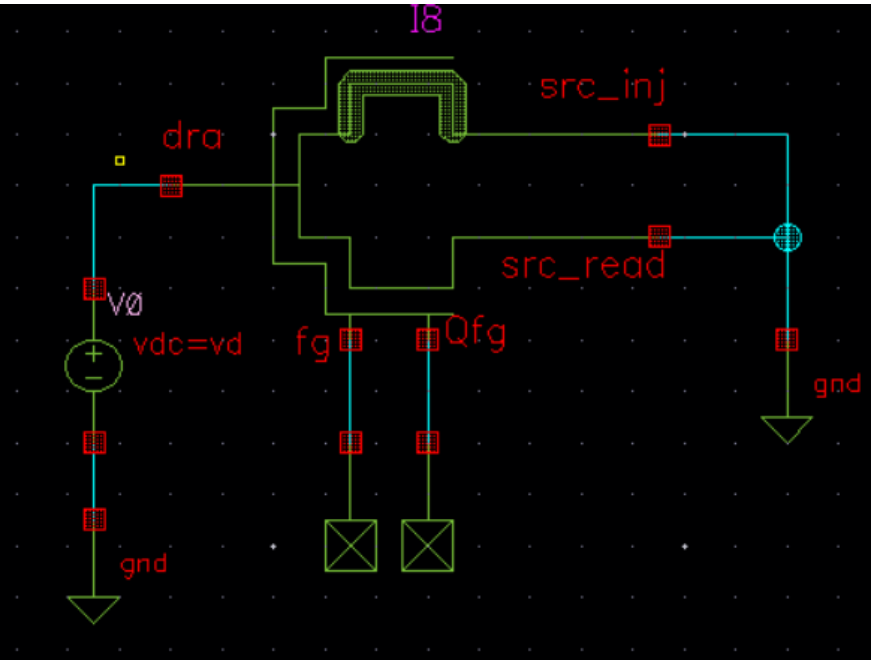
    //parameters for readout
    parameter real Vth   = 0.85;
    parameter real K     = 1.2e-5;
    parameter real Is0   = 2e-8;
    parameter real n     = 1.7;
    //parameters for channel hot electron current
    parameter real Prob0 = 3.9e-8;
    parameter real Va    = 20;
    //parameters for tunneling current
    parameter real beta  = 10;
    parameter real Vbi   = 5.5;
    parameter real xi    = 2.0e-12;
    parameter real kT = $vt;

    //internal variables
    real Vds, Vgs;
    real Vsg;
    real is,is_sub,is_ab,igs_t,igs_1,igs; //current
    real m=1, s=5;

    analog begin
        // current of the readout transistor
        if (V(dra,gnd) == V(src,gnd))
            is = 0;
        else begin
            Vds = abs(V(dra)-V(src));
            if (V(dra)>V(src))
                Vgs=V(g)-V(src);
            else
                Vgs=V(g)-V(dra);
            is_sub = Is0*exp((Vgs-Vth)/(n*kT))*(1-exp(-Vds/kT));
            is_ab = s*Is0*(1-exp(-Vds/kT));
            if (Vgs-Vth >= 0) begin
                if (V(g)-Vth < Vds)
                    is_ab = is_ab + K/2*(Vgs - Vth)**2;
                else
                    is_ab = is_ab + K*(Vgs - Vth-Vds)*Vds;
            end
            is = (1/(1/(is_sub**m)+1/(is_ab**m)))**(1/m);
            if (V(dra)<V(src))
                is=-is;
            end
            // lucky electron gate current
            if (V(g)<0.1)
                igs_1 = 0;
            else
                igs_1=abs(is)*Prob0*exp(-Va/V(g));
            // tunneling gate current
            Vsg=V(src)-V(g);
            if (Vsg<Vbi)
                igs_t = 0;
            else
                igs_t = -xi*(Vsg-Vbi)**2*exp(-beta/(Vsg-Vbi));
            // total gate current in the readout transistor
            igs=igs_1+igs_t;

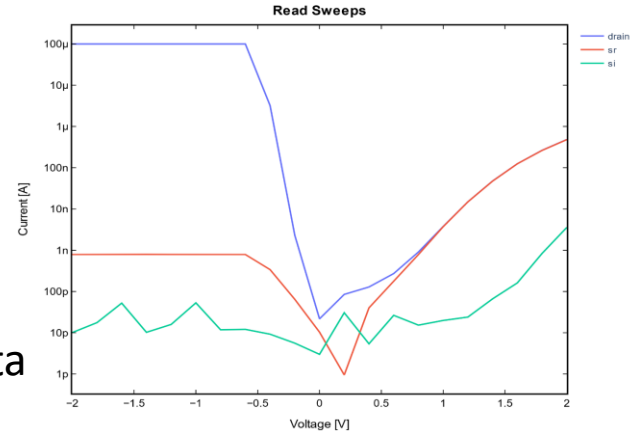
            I(dra,src) <+ is;
            I(g,src) <+ igs;
        end
endmodule
```

➢ Verilog-a Model Results - 1

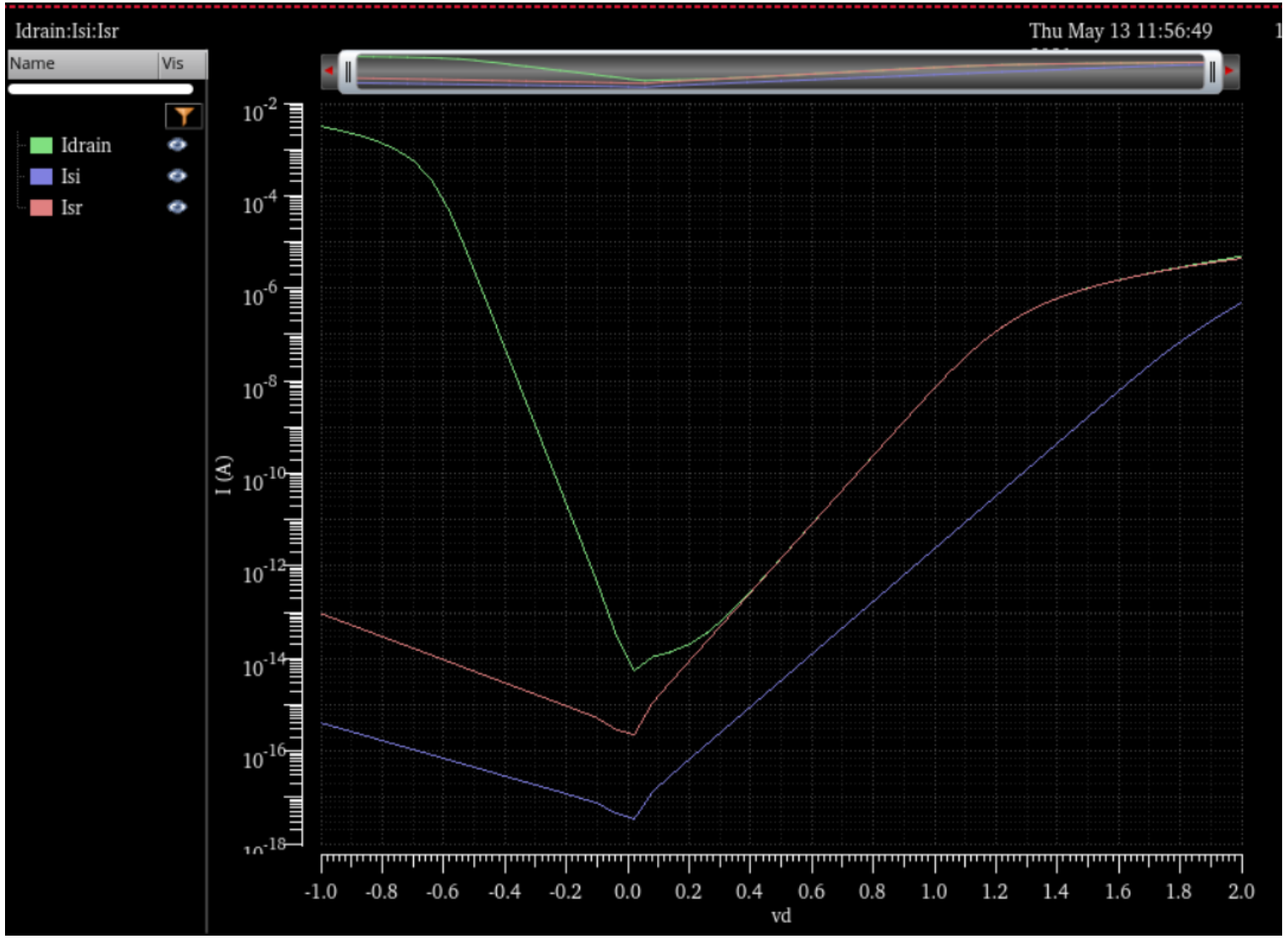


Qfg = 0

- Data

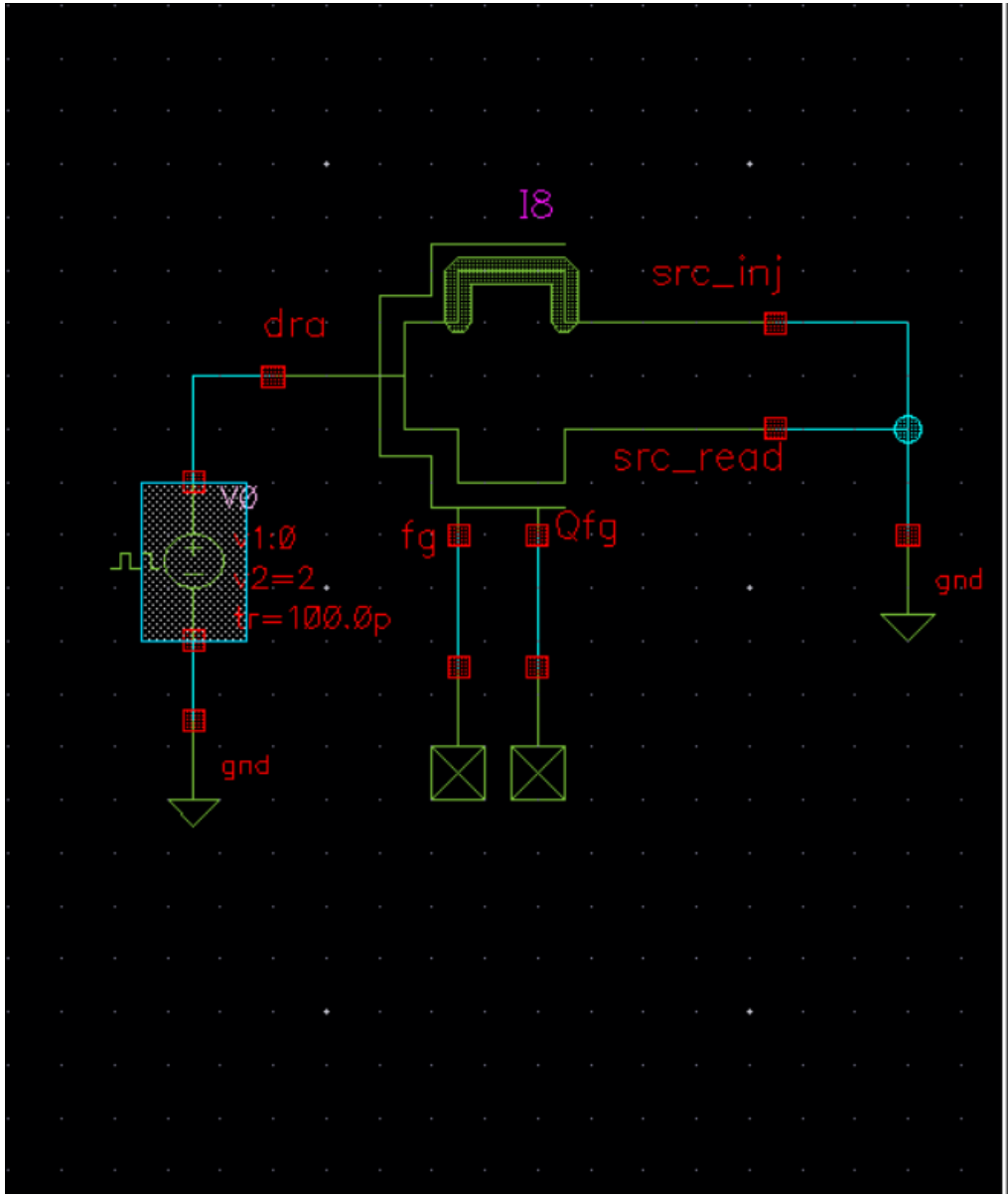


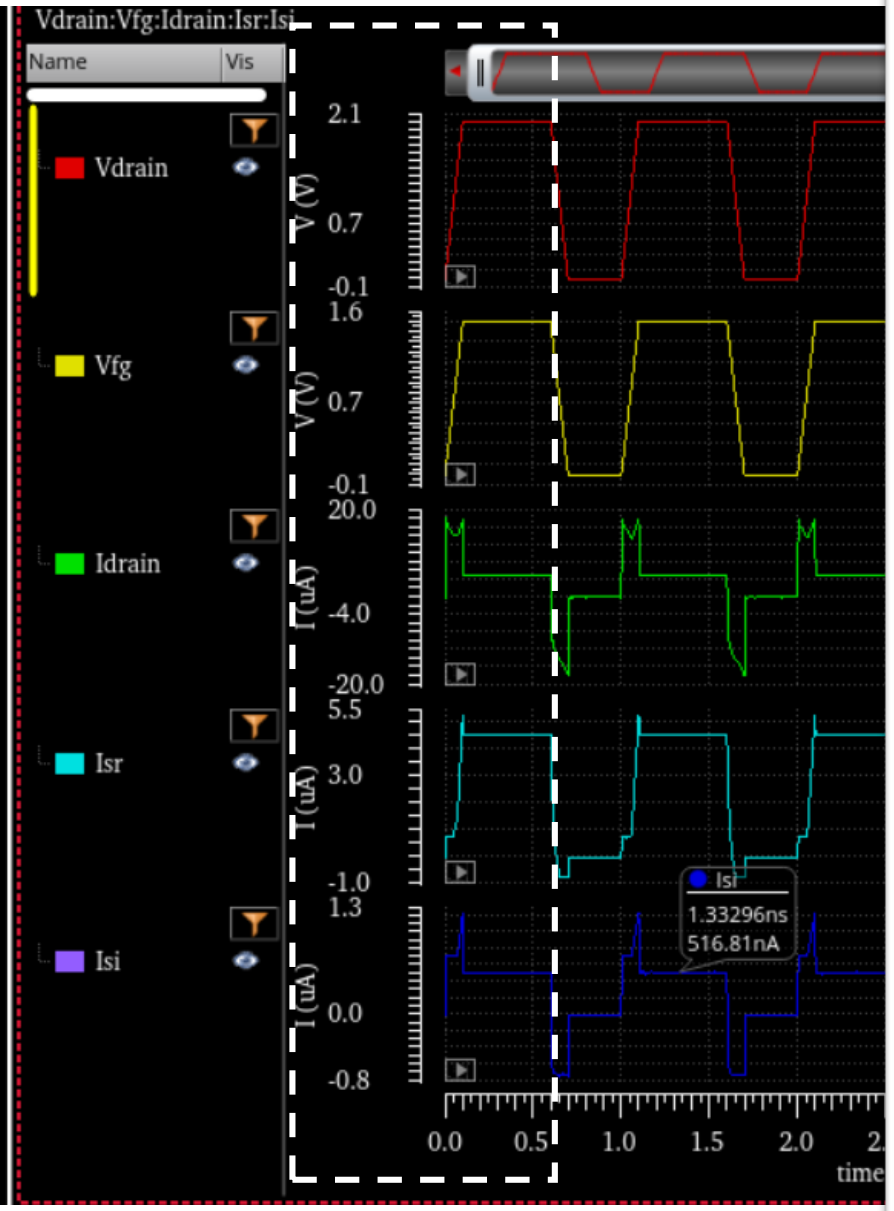
- DC Reading

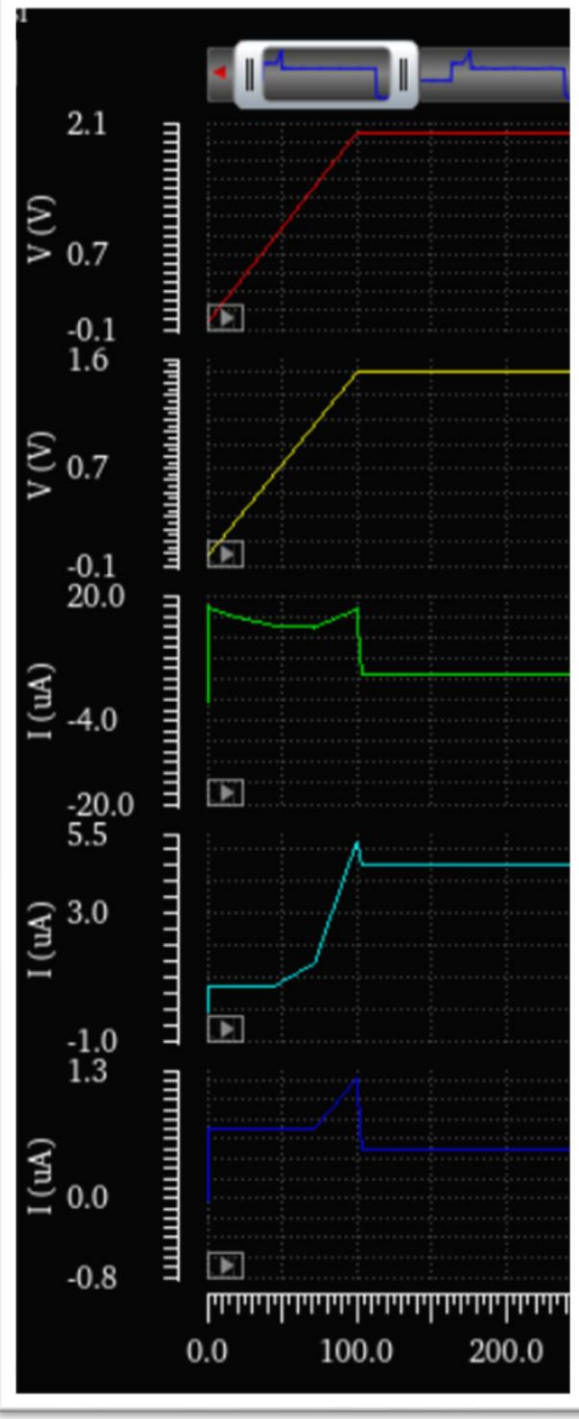


➢ Verilog-a Model Results - 2

- Pulse Reading, tr=0.1ns

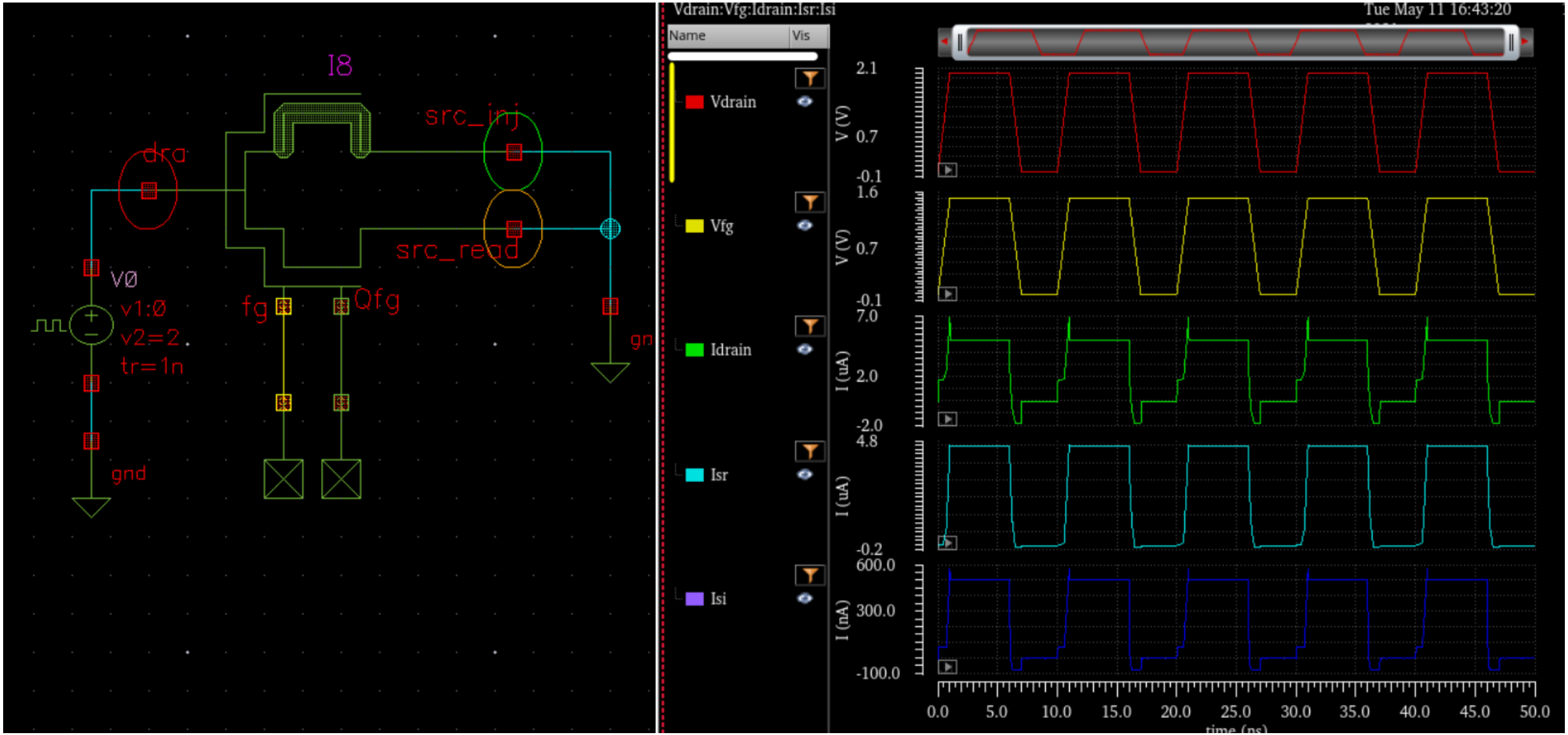






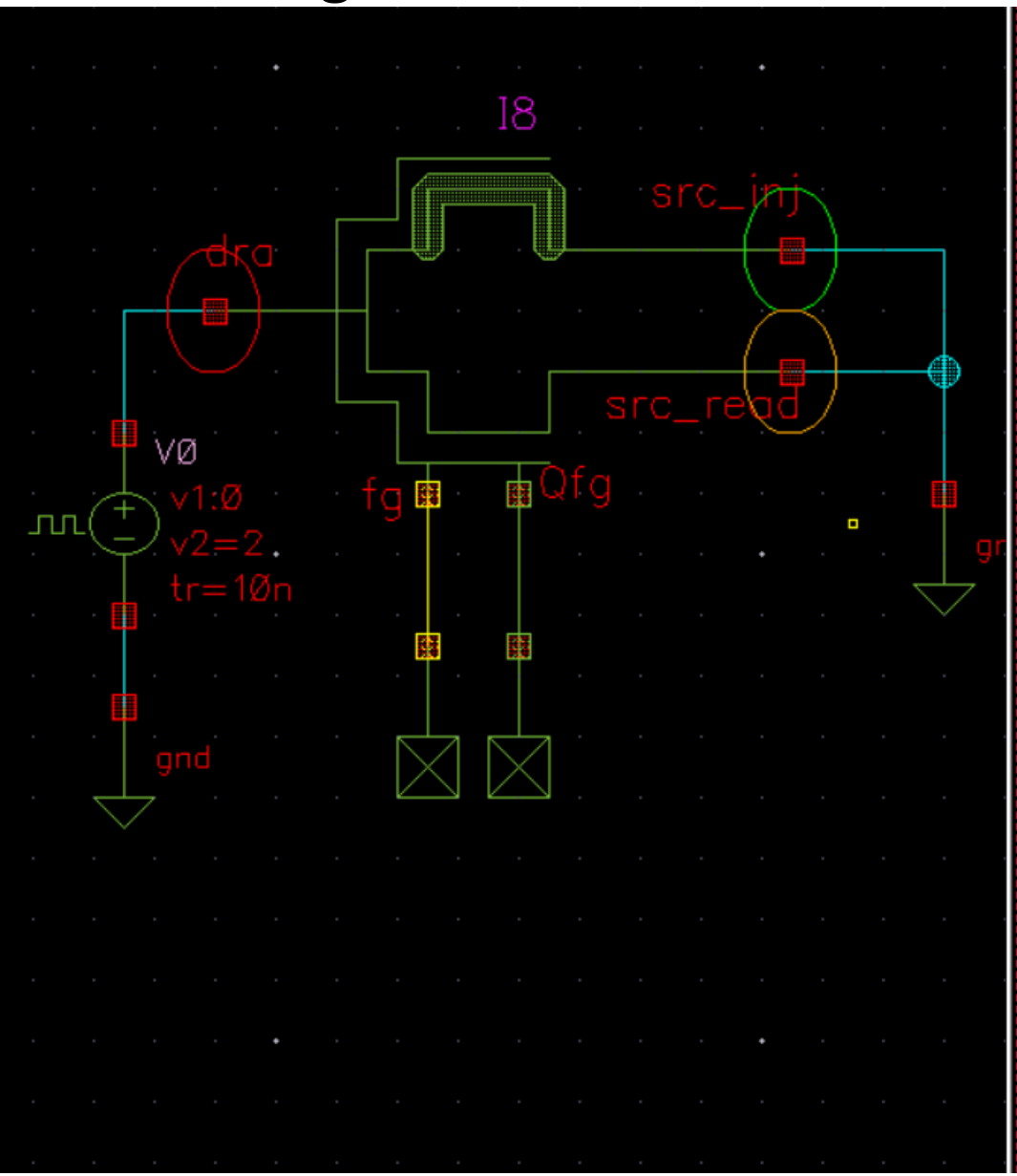
➢ Verilog-a Model Results - 3

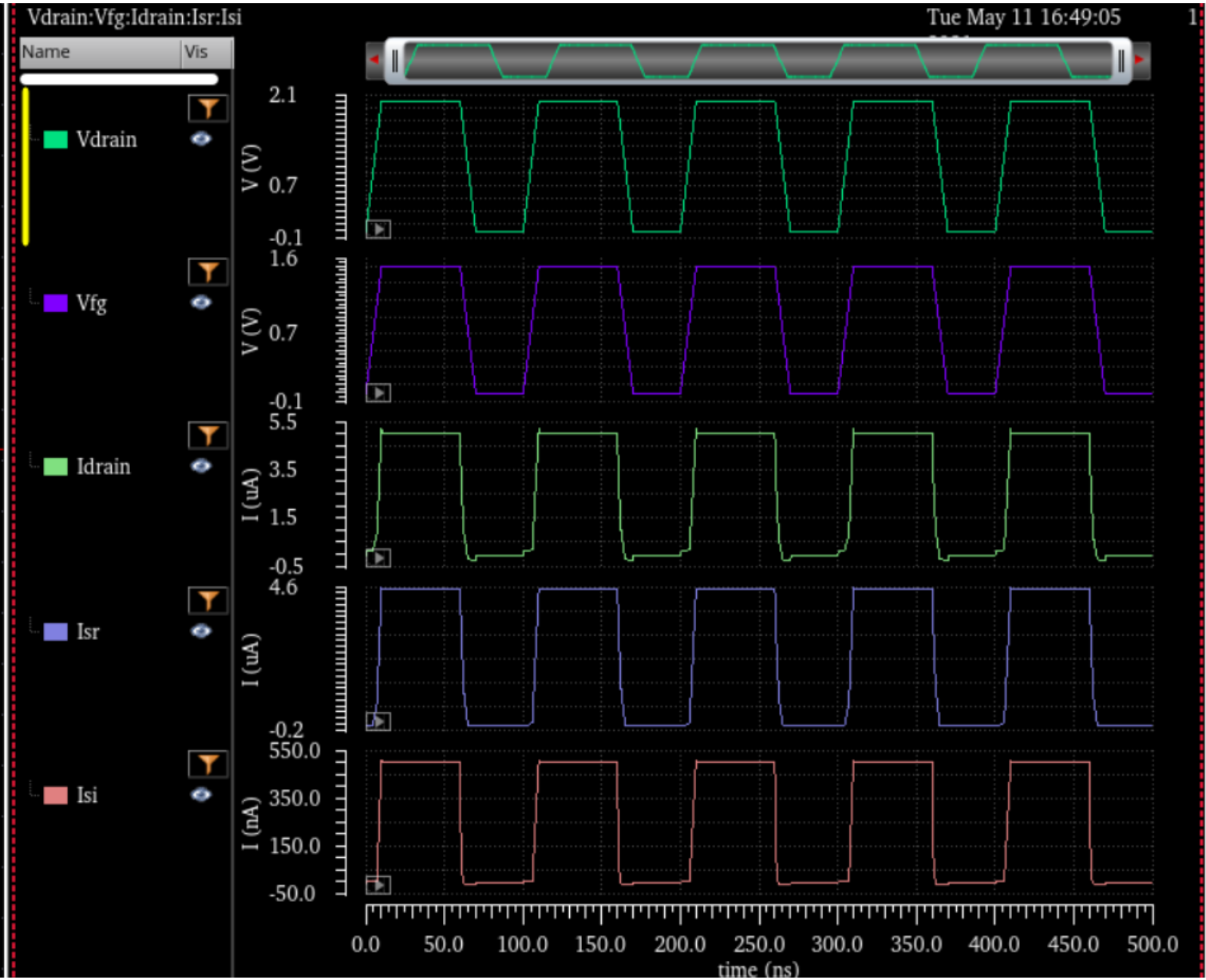
- Pulse Reading, tr=1ns



➢ Verilog-a Model Results - 4

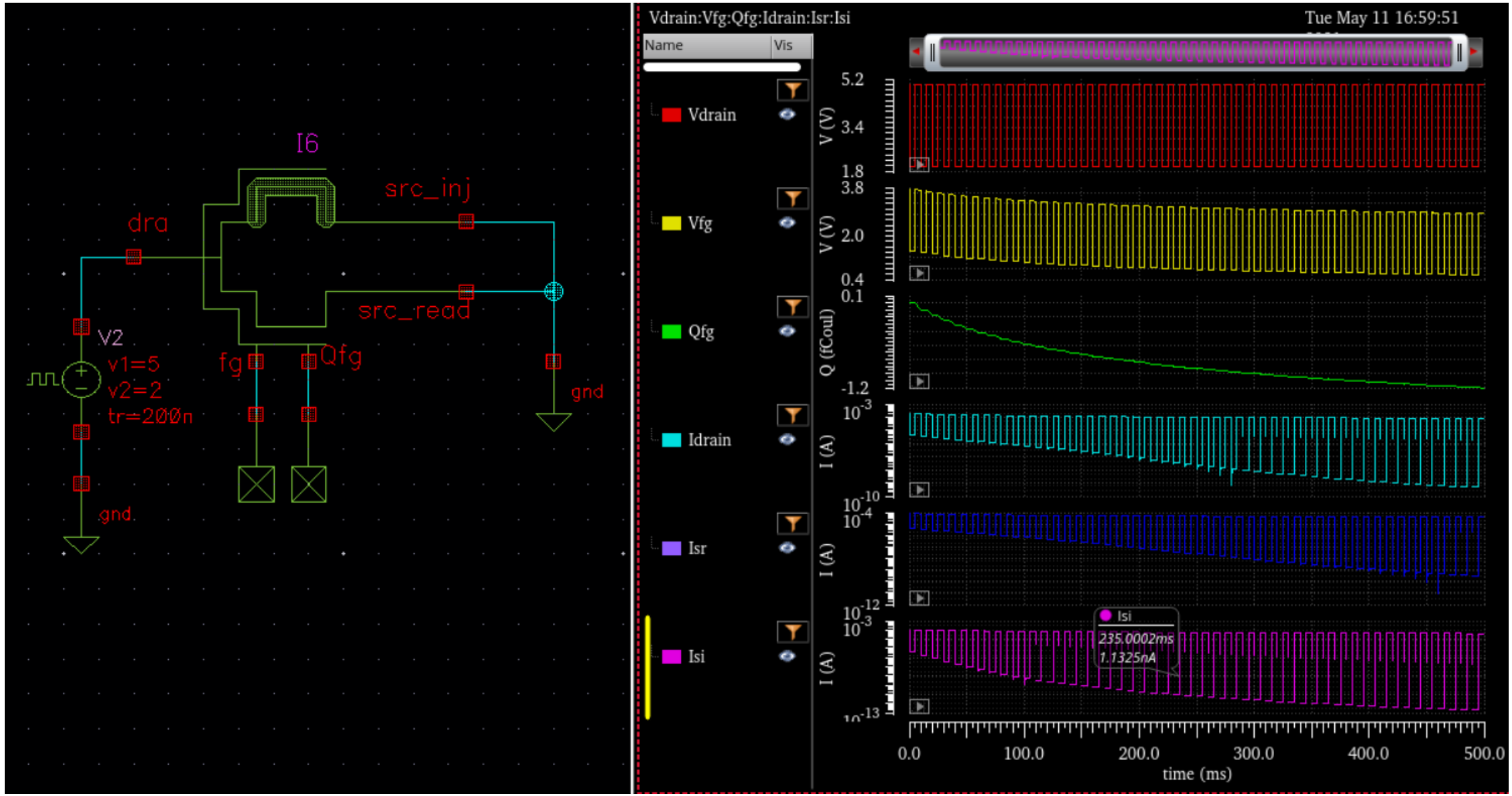
- Pulse Reading, tr=10ns





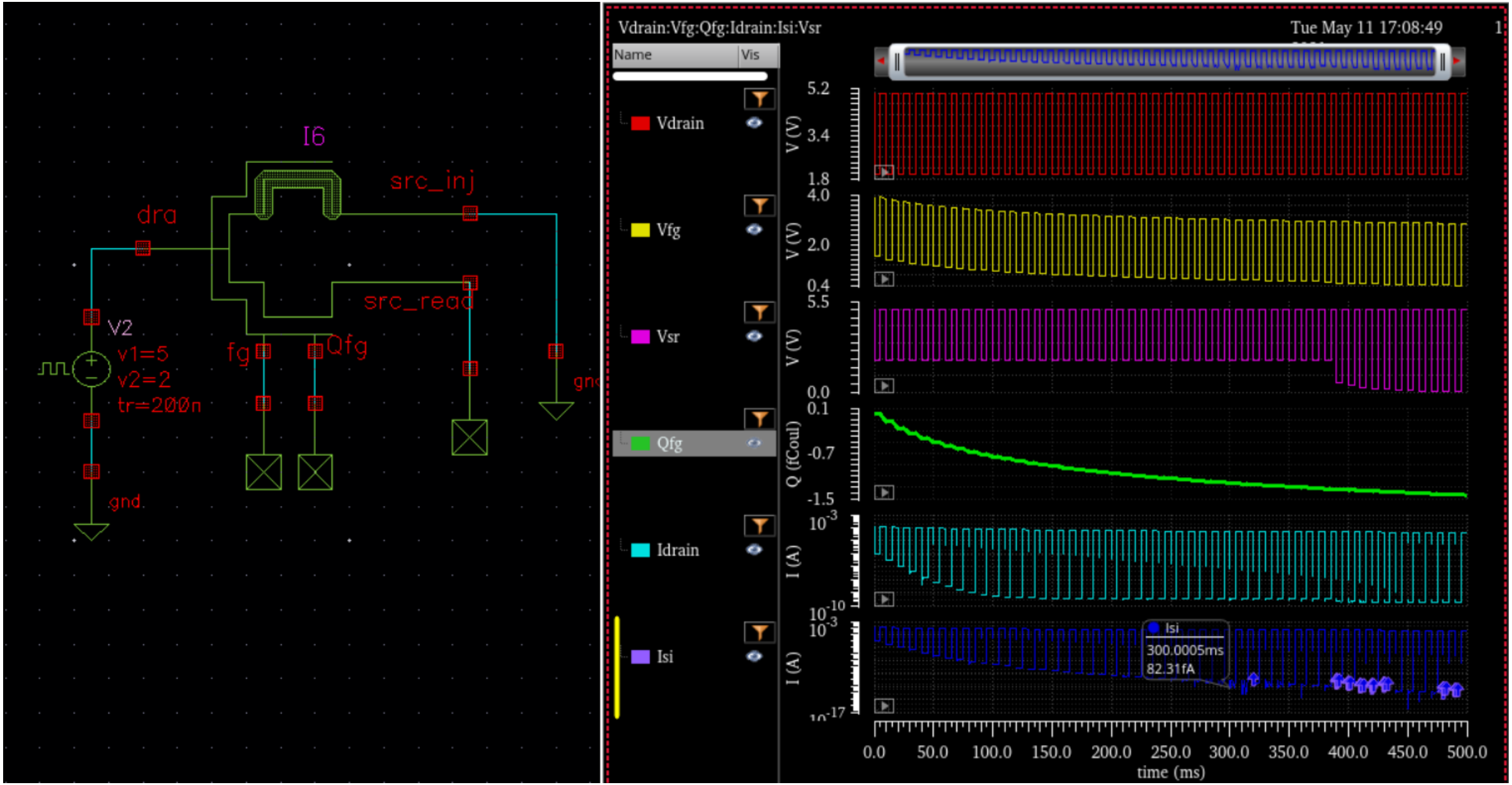
➢ Verilog-a Model Results - 5

- Program, SR shorted with SI



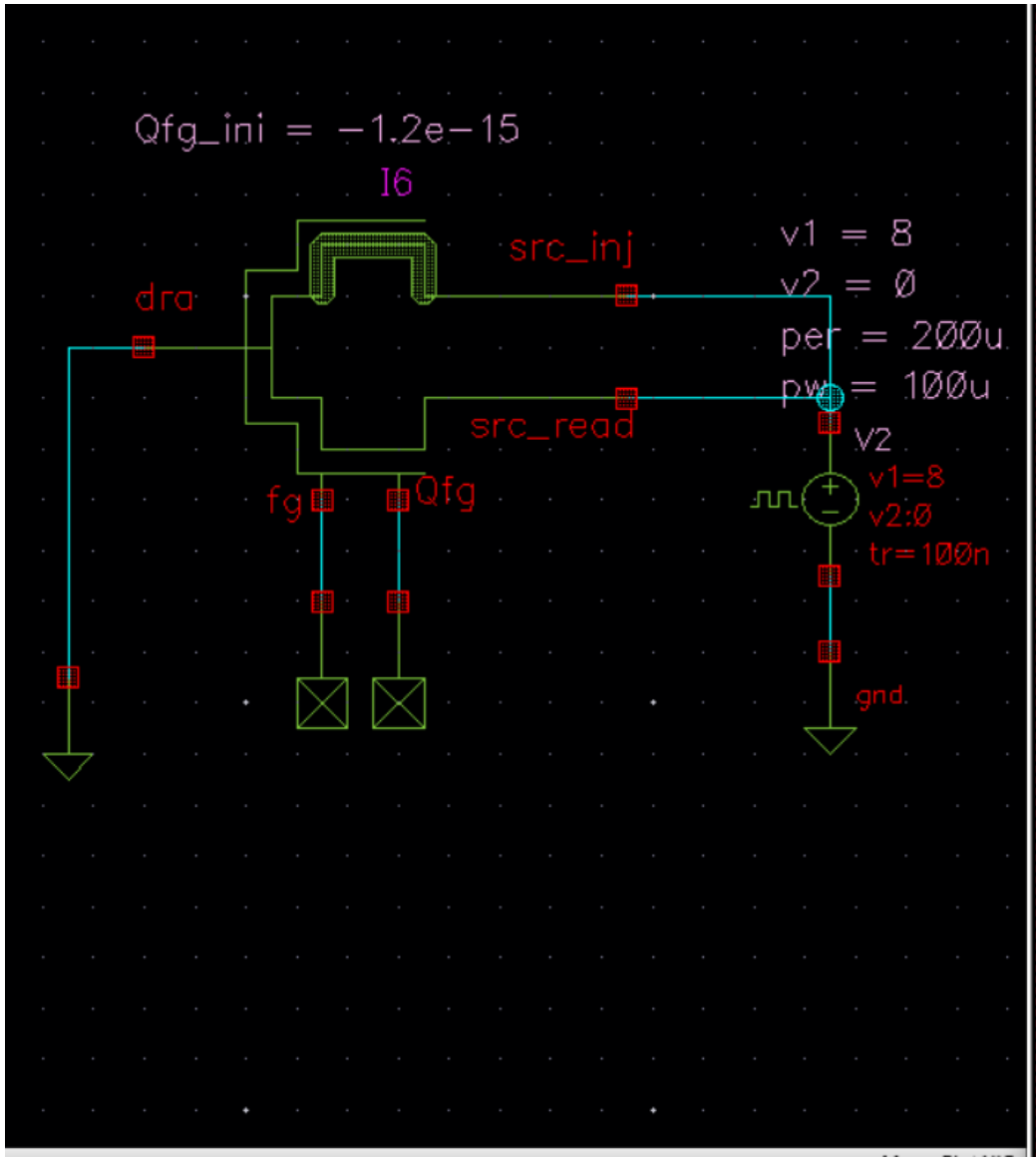
➢ Verilog-a Model Results - 6

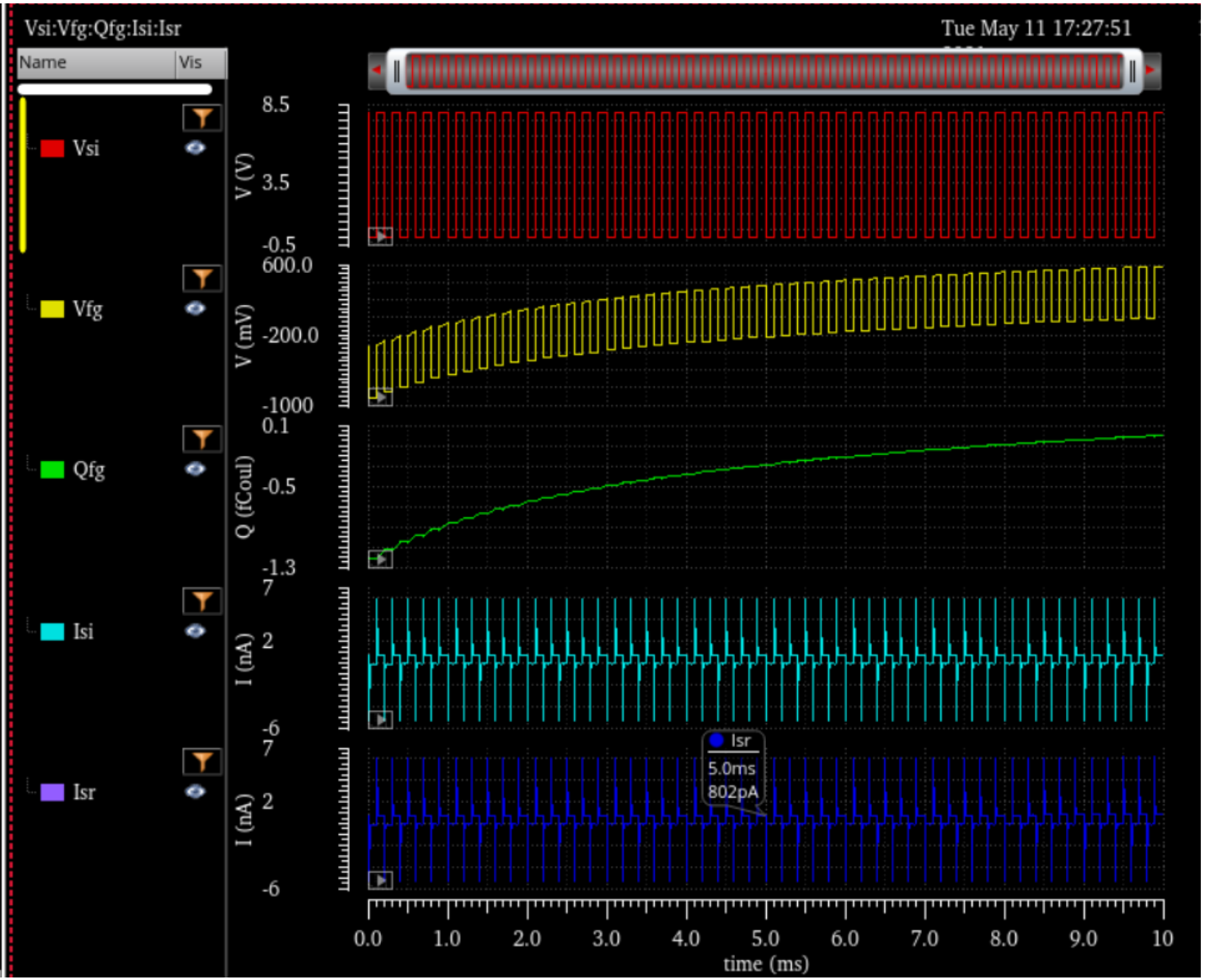
- Program, SR floating



➢ Verilog-a Model Results - 7

- Erase, SR shorted to SI





➢ Verilog-a Model Results - 8

- Erase, SR floating

